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Simulation of Accelerator-Driven Transmutation of the U.S. Spent Nuclear Fuel Inventory

by

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Thesis

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Abstract

The transmutation of minor actinides in an accelerator driven subcritical system (ADS) is often proposed as a means of reducing the long term radiological hazard of used nuclear fuel. Evaluating such a mission at the national scale requires connecting a realistic description of the existing spent fuel inventory to a model of the burner that would consume it. This thesis develops two facility archetypes for the Cyclus fuel cycle simulator to make that connection. The first, `us_inventory`, represents the U.S. inventory of used fuel at the resolution of individual assemblies, drawn from the STANDARDS 5.0.1 dataset, and supplies material to downstream facilities according to a choice of ranking policies while preserving each assembly's composition and geographic provenance. The second, `ads`, models an accelerator driven minor actinide burner using precomputed Serpent depletion cases as a reduced order model, transforming a loaded core into its irradiated composition and recording the actinide mass destroyed each cycle.

Using these archetypes, a transmutation fuel cycle is simulated for two burner configurations, a minor actinide only core and a fifty-fifty plutonium minor actinide core, over 30- and 100-year horizons in once through and recycling modes. The results are governed by the scarcity of the feed: the accessible minor actinide inventory is only about 104 t, fewer than four loadings of the large minor actinide core. As a result, the minor actinide burner is feed limited, stalling after three cores without recycling and five with it, whereas the smaller fifty-fifty core, sustained by the far more abundant plutonium, runs continuously across the simulation. The transmutation mission is therefore constrained by how quickly its dilute feed can be retrieved, separated, and recycled rather than by installed reactor power, and because minor actinides are only about 0.1% of the spent fuel, the mission changes the isotopic character of the waste without reducing the mass sent to disposal. The archetypes and the assembly resolved, provenance preserving results they produce provide a foundation for subsequent radiological, economic, and siting analyses of accelerator driven transmutation.

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List of Abbreviations

AAA Advanced Accelerator Applications.

ADS accelerator-driven system.

ANICCA Analysis of Nuclear-fuel-cycles Including Chain and Cost Assessment.

API application programming interface.

ARPA-E Advanced Research Projects Agency-Energy.

ATW Accelerator Transmutation of Waste.

BWR Boiling Water Reactor.

CEA Commissariat à l'Énergie Atomique et aux Énergies Alternatives.

CLEAR China Lead-based Reactor.

CNRS National Centre for Scientific Research.

COSI Commelini-Sicard.

DOE Department of Energy.

DRE dynamic resource exchange.

EFIT European Facility for Industrial Transmutation.

EINSTEIN Enhanced Integrated Nuclear Systems for Transmutation and Efficient Isolation of Nuclides.

FDS fusion-driven system.

GENEPI-3C Générateur de Neutrons Pulsés Intenses, third generation with continuous beam.

GUINEVERE Generator of Uninterrupted Intense Neutrons at the Lead VENUS Reactor.

ISFSI Independent Spent Fuel Storage Installation.

LWR Light Water Reactor.

MEGAPIE MEGAwatt Pilot Experiment.

MINERVA MYRRHA Injector and low-energy accelerator.

MTHM metric tons of heavy metal.

MTU metric tons of uranium.

MYRRHA Multi-purpose Hybrid Research Reactor for High-tech Applications.

NEA Nuclear Energy Agency.

NEWTON Nuclear Energy Waste Transmutation Optimized Now.

NGSAM Next-Generation System Analysis Model.

OECD Organisation for Economic Co-operation and Development.

.

PWR Pressurized Water Reactor.

SCK CEN Belgian Nuclear Research Centre.

SINQ Swiss Spallation Neutron Source.

SNF spent nuclear fuel.

STANDARDS Storage Transportation and Disposal Analysis Resource and Data System.

UDB Unified Database.

UIUC University of Illinois Urbana-Champaign.

UNF used nuclear fuel.

VENUS-F fast-spectrum configuration of the Vulcan Experimental Nuclear Study reactor.

XDS accelerator- or fusion-driven system.

List of Symbols

i	Mass-number index used when summing isotopes of an element.
$a_c(t)$	Actinide mass fraction of depletion case c at time t .
c	Index identifying a Serpent depletion case.
$\mathcal{E}$	Set of inventory bins eligible to supply a request.
$f_{\text{fissile},a}$	Fissile mass fraction of assembly a .
$f_{\text{MA},a}$	Minor-actinide mass fraction of assembly a .
$f_{\text{Pu},a}$	Plutonium mass fraction of assembly a .
$f_{\text{MA}}, f_{\text{Pu}}, f_{\text{U}}$	Minor-actinide, plutonium, and uranium fractions of an ADS core, normalized to total actinide mass.
$f_g, f_{g,c}$	Fraction of actinide group g in the loaded core and in depletion case c , respectively.
g	Actinide-group index, where $g \in \{\text{MA}, \text{Pu}, \text{U}\}$.
a	Assembly index.
k_{eff}	Effective neutron multiplication factor.
m_a	Total or remaining mass of assembly a .
$m_{a,t}$	Remaining mass of assembly a at simulation time step t .
$m_{i,a}$	Mass of nuclide i in assembly a .
$m_n(t)$	Mass of nuclide n in a depletion case at irradiation time t .
$m_{\text{Np}}, m_{\text{Am}}, m_{\text{Cm}}$	Total masses of neptunium, americium, and curium in a loaded core.
$m_{\text{Pu}}, m_{\text{U}}$	Total plutonium and uranium masses in a loaded core.
m_{act}	Total actinide mass in a loaded core.
m_{supplied}	Mass delivered in one inventory trade.
$M(t)$	Total tracked mass in a depletion case at irradiation time t .
M_{eligible}	Total mass in the eligible inventory pool.
$M_{\text{throughput}}$	Configured maximum inventory throughput per time step.
M_{offer}	Maximum mass offered by the inventory facility in a time step.

$M_{\text{inv},t}$	Total mass remaining in the inventory facility at time step t .
M_{in}	Core mass entering the ADS composition transformation.
M_{out}	Discharged mass after the ADS composition transformation.
n_{cycle}	ADS irradiation-cycle length in Cyclus time steps.
P, P_c	ADS facility power and power associated with depletion case c .
R_t	Inventory throughput remaining during time step t .
R	Ratio of discharged tracked mass to initially tracked mass in the selected depletion case.
S_c	Matching score assigned to depletion case c .
t	Time or simulation-time index, as defined locally.
t_{irr}	Irradiation time corresponding to an ADS discharge.
$t_{\text{lo}}, t_{\text{hi}}$	Tabulated depletion times immediately below and above t_{irr} .
t_k	A tabulated time point in a Serpent depletion case.
w_P	Weight applied to the power term in ADS case matching.
$\mathbf{x}_i$	Nuclide mass-fraction vector of assembly i .
$\mathbf{x}_{\text{bid}}$	Mass-weighted composition vector used for an inventory bid.
X	Generic nuclide identifier.
Z	Atomic number.
α	Linear-interpolation fraction between t_{lo} and t_{hi} .
Δ_d	Number of irradiation days represented by one Cyclus time step.
ΔM_{act}	Actinide mass destroyed during one ADS cycle.

Chapter 1

Introduction

Nuclear power has supplied the United States with large quantities of low-carbon electricity for more than six decades. The fuel discharged from those reactors, however, remains a long term national responsibility. By the end of 2025, commercial reactors in the United States had permanently discharged roughly 95,000 metric tons of heavy metal (MTHM) of used nuclear fuel (UNF), and the inventory continues to grow by approximately 2,000 MTHM each year [1,2]. Extrapolating this rate places the inventory at approximately 99,000 MTHM in 2026. This material is distributed across more than one hundred sites in thirty-nine states, including operating nuclear power plants, decommissioned reactor sites, and independent spent fuel storage installations. Because the United States does not yet operate a permanent geologic repository for commercial spent fuel, most of the inventory remains stored at or near the sites where it was generated [3]. The challenge is not simply that the inventory is large. It is also highly heterogeneous. The phrase “the U.S. used nuclear fuel inventory” can suggest a single stockpile with one representative composition, but the actual inventory is the accumulated result of decades of reactor operation. Assemblies discharged in different years were fabricated with different initial enrichments, irradiated to different burnups, operated in different reactor types, and cooled for different lengths of time. Their remaining heavy metal masses and isotopic compositions therefore vary considerably. Some assemblies contain larger quantities of residual fissile material, plutonium, or minor actinides than others. Some have cooled for several decades, while others were discharged recently and continue to generate substantial decay heat. Any strategy that attempts to process, recycle, or transmute this material must eventually confront those differences.

UNF also changes with time. During the first decades after discharge, its heat generation and radiation field are strongly influenced by fission products. As those nuclides decay, the relative importance of plutonium and the minor actinides increases. Their long half-lives contribute to the need to isolate untreated fuel over timescales on the order of 10^5 years [4]. Among the long-lived components of used fuel, the actinides are distinctive because they can be destroyed through fission rather than managed only through continued storage and radioactive decay. That possibility does not eliminate the need for a repository, nor does it remove every radioactive waste stream. It does, however, create the possibility of changing the quantity and composition of the material that must ultimately be isolated.

Accelerator-driven systems (ADSs) have been proposed as one technology for carrying out this transmutation. An ADS couples an externally generated neutron source to a subcritical multiplying core. Because the core remains below criticality, its operation depends on the external source, allowing fuel compositions with high concentrations of minor actinides to be considered under conditions that would be more difficult

to accommodate in a critical reactor. Detailed studies have examined accelerator reliability, spallation targets, subcritical core physics, liquid metal coolants, fuel behavior, and the rates at which plutonium and minor actinides could be consumed. These studies establish the physical basis for accelerator driven transmutation and provide increasingly realistic designs for future facilities.

A technically feasible core, however, is not the same as a viable transmutation strategy. A reactor physics calculation may show how much actinide material one system can consume, but it does not determine where that material will come from, which assemblies will be processed first, how the feed will change over time, or how many facilities will be required. Before an ADS can consume actinides, spent fuel must be selected from storage, transported, cooled as needed, separated, and converted into a suitable feed form. The transmutation facility must then be deployed, supplied, operated, and eventually integrated with storage, recycle, and waste management facilities. The material discharged from the ADS may require further processing or repeated recycle before the desired degree of actinide destruction is achieved.

The central problem is therefore a fuel cycle problem rather than only a single core problem. It requires a model that can follow material from the existing inventory, through selection and processing, into a transmutation facility, and finally into the inventories and waste streams that remain. Such a model must operate over timescales of years or decades while retaining enough information to distinguish among assemblies. It must also allow facilities to compete for resources, enforce throughput limits, and respond to changes in material availability. These capabilities are difficult to represent using a single average spent fuel composition.

Many previous fuel cycle studies have necessarily simplified the inventory by representing it as an aggregated material stream or a small set of representative recipes [5–7]. That approach is useful when comparing generic future fuel cycles or studying equilibrium reactor fleets. It is less satisfactory when the starting point is the existing U.S. inventory. A national average recipe cannot identify whether the material it represents comes from older or newer assemblies, whether it has high or low burnup, whether it is concentrated at a few sites or dispersed across the country, or which assemblies would actually be selected under a particular management strategy. These differences may affect the amount and quality of material available for transmutation and the characteristics of the inventory that remains untreated.

Recent inventory databases make a more realistic representation possible. The Storage Transportation and Disposal Analysis Resource and Data System (STANDARDS) 5.0.1 database [8,9] contains assembly-level information describing the U.S. commercial spent fuel inventory, including attributes such as discharge date, burnup, initial enrichment, mass, and calculated isotopic composition. This level of detail creates an opportunity to treat the national inventory not as a single homogeneous stockpile, but as a collection of discrete resources that can be selected according to explicit criteria. The database alone, however, does not describe how those assemblies would move through a future fuel cycle. The information must be incorporated into a dynamic simulator capable of representing transactions among facilities.

The work is conducted within the Advanced Reactors and Fuel Cycles group at University of Illinois Urbana-Champaign (UIUC), where agent-based fuel cycle simulation is used to examine the system level consequences of advanced nuclear technologies. It also contributes to the UIUC led Enhanced Integrated Nuclear Systems for Transmutation and Efficient Isolation of Nuclides (EINSTEIN) project under the Advanced Research Projects Agency-Energy (ARPA-E) Nuclear Energy Waste Transmutation Optimized Now (NEWTON) program, which brings together neutronics, nuclear data, materials, fuel cycle, and techno-economic analyses to evaluate externally driven transmutation systems. The topic of this thesis involves simulation with CYCLUS, detailed spent fuel inventory data, and higher fidelity reactor calculations,

to fill a gap between the available inventory information and existing system level transmutation models. This thesis connects the fuel assemblies that exist today with the transmutation technologies being proposed for the future.

It uses the CYCLUS nuclear fuel cycle simulation framework to make that connection. CYCLUS represents the fuel cycle as a system of interacting agents, including regions, institutions, and facilities [10]. Individual facilities request, offer, and exchange material through the Dynamic Resource Exchange [11], while modular software components known as archetypes define their behavior. This structure makes it possible to represent the U.S. inventory as an active supplier rather than as a static input file [12]. It also allows that supplier to interact with processing, storage, and transmutation facilities within the same simulation. Two new facility archetypes are developed for this purpose. The `us_inventory` archetype represents individual U.S. spent fuel assemblies as selectable material resources. Each assembly retains metadata including its heavy metal mass, discharge date, burnup, initial enrichment, and isotopic composition derived from STANDARDS 5.0.1. Rather than collapsing these records into one average composition, the facility supplies material through the CYCLUS Dynamic Resource Exchange according to user defined throughput limits and inventory selection policies.

The second archetype `ads`, represents an ADS. Its material transformation is based on precomputed Serpent depletion calculations, allowing the fuel cycle simulation to incorporate higher fidelity neutronics results without performing a full transport and depletion calculation during every simulation time step. Coupling the two archetypes makes it possible to follow material from individual inventory records into the modeled transmutation system and to observe how the remaining inventory changes as material is withdrawn.

The selection policies considered in this work rank available assemblies according to characteristics such as discharge date, burnup, and initial enrichment. These policies are not intended to represent final operational decisions or recommendations for U.S. waste management. Instead, they provide controlled cases for examining how different assumptions about access to the inventory affect the material supplied to a transmutation system. For example, prioritizing older assemblies may produce a different feed composition than prioritizing high burnup or high enrichment assemblies, even when the same total mass is supplied. The simulation records these differences and preserves the connection between the supplied material and the assemblies from which it originated.

The primary contribution of this thesis is a material flow model rather than a new accelerator driven reactor design or a full evaluation of whether the United States should deploy accelerator driven transmutation. A decision of that kind would still require detailed neutronics, thermal hydraulics, materials, safety, economic, transportation, safeguards, and repository analyses. What this work provides instead is a model that connects a realistic, assembly resolved representation of the U.S. inventory to an ADS facility inside a dynamic fuel cycle simulation.

This connection provides a foundation for broader policy and deployment studies. A future analysis could add site specific transportation, separations capacity, construction schedules, facility siting, operating costs, or alternative transmutation technologies. Those questions cannot be fully grounded in the present inventory, however, unless the model first represents the assemblies that actually exist and the rules by which they become available. By preserving assembly-level differences, this work makes the assumptions behind a transmutation scenario more transparent and its results more traceable.

The broader motivation is to ground the discussion of transmutation in how a real national inventory could supply such facilities over time, rather than in what a single idealized facility can destroy on its

own. This emphasis matters because the performance of a technology in isolation may differ from its performance as part of an operating fuel cycle. Through the integration of assembly-level inventory data and an ADS representation in CYCLUS, this thesis provides one step toward evaluating that larger system. The remainder of this thesis is organized as follows. Chapter 2 reviews UNF management in the United States, the CYCLUS framework, accelerator driven systems, the EINSTEIN program, and prior transmutation fuel cycle simulations. It concludes by identifying the research gap addressed in this work. Chapter 3 describes the design and implementation of the `us_inventory` and `ads` archetypes, the preparation of their input data, and the simulation scenarios considered. Chapter 4 presents the simulation results and compares the effects of the alternative inventory selection policies. Finally, Chapter 5 discusses the implications and limitations of the results and identifies directions for future development.

Chapter 2

Background and Literature Review

This chapter establishes the technical and programmatic background on which the remainder of the thesis builds. The problem introduced in Chapter 1, that a large, heterogeneous, and geographically distributed inventory of UNF must eventually be managed rather than only stored, draws together several areas that have historically been developed in isolation: the characteristics of the national inventory itself, the simulation tools used to model fuel cycles, the transmutation technology proposed to consume long-lived actinides, and the research program within which this work is situated. Each is reviewed in turn. Section 2.1 describes the scale, distribution, and composition of UNF in the United States and the assembly-level data that characterize it. Section 2.2 introduces the CYCLUS agent-based fuel cycle simulation framework and the modeling capabilities on which the archetypes developed in this work depend. Section 2.3 presents accelerator-driven systems and their role in minor actinide transmutation. Section 2.4 situates the thesis within the EINSTEIN project and the broader ARPA-E NEWTON program. Section 2.5 reviews prior fuel cycle simulations of transmutation systems, and Section 2.6 draws these threads together to identify the specific research gap that this thesis addresses.

2.1 Used Nuclear Fuel in the United States

UNF, also commonly called spent nuclear fuel, is nuclear fuel that has been removed from a reactor because it can no longer efficiently support the desired operating cycle. Although the fuel is no longer used in the reactor, it remains thermally hot and highly radioactive because it contains radioactive fission products, plutonium, and minor actinides produced during irradiation [13]. The term “used” is particularly appropriate because the discharged material is not completely depleted: most of its uranium remains, and the fuel also contains newly generated fissile isotopes that could potentially be recovered and reused. The radioactivity and heat generation of UNF decrease with time, but they do so over multiple timescales. Short-lived fission products dominate the radiation field and decay heat immediately following discharge. At longer times, plutonium and the minor actinides become increasingly important because many of these isotopes have half-lives ranging from hundreds to tens of thousands of years. Consequently, untreated commercial UNF may require isolation over timescales on the order of 10^5 years before its radiotoxicity approaches that of the natural uranium ore from which the fuel was originally produced [4]. The long duration of this hazard is a central challenge in nuclear waste management.

The internationally accepted approach for the permanent isolation of UNF and high-level radioactive

waste is disposal in a deep geologic repository [14]. The United States, however, does not currently operate a permanent repository for commercial UNF. Until such a repository or another disposition pathway becomes available, reactor licensees must store the fuel at or near the sites where it was generated [3,13]. Newly discharged fuel is first placed in water filled spent fuel pools, which provide both cooling and radiation shielding. After several years of cooling, assemblies may be transferred to sealed dry storage systems at independent spent fuel storage installations.

Because a national disposal system has not yet been implemented [3], the geographic distribution of UNF largely reflects the historical deployment of the U.S. reactor fleet. Most commercial fuel remains at operating or decommissioned nuclear power plant sites, including locations at which the reactor itself has been permanently shut down. The result is a distributed inventory that must eventually be transported, consolidated, processed, or disposed of [2].

Approximately 99,000 MTHM from commercial pressurized water reactors and boiling water reactors had been permanently discharged in the United States. This inventory grows by roughly 2,000 MTHM per year as operating reactors discharge fuel [15], and the current reactor fleet is projected to have generated about 140,000 MTHM in total by 2060. The U.S. Department of Energy (DOE) managed an additional inventory of approximately 2,446 MTHM, excluding most naval fuel. Unlike the comparatively standardized commercial light water reactor inventory, DOE-managed UNF includes fuel from defense production reactors, research and test reactors, early demonstration reactors, and other specialized programs. Altogether, UNF and primary reprocessing wastes were distributed across more than 100 locations in 39 states [2].

Table 2.1, compiled from the PNNL *Spent Nuclear Fuel and Reprocessing Waste Inventory* report [2] and the GC-859 Nuclear Fuel Data Survey [15], summarizes the scale and distribution of this inventory. Commercial fuel dominates the national inventory by mass, while DOE-managed material is considerably more diverse in enrichment, fuel form, cladding, geometry, and irradiation history.

Table 2.1: The U.S. inventory of used nuclear fuel and primary reprocessing waste was distributed among commercial, DOE, research, and legacy waste sites. Data from [2,15].

Inventory category	Reported quantity
Commercial UNF at nuclear power plant and ISFSI sites	99,000 MTHM
Permanently discharged PWR fuel	135,645 assemblies; 59,013 MTHM
Permanently discharged BWR fuel	179,466 assemblies; 32,023 MTHM
Commercial dry storage canisters or casks	3,862
DOE-managed UNF	approximately 2,446 MTHM ^a
UNF at university, government, and commercial research sites	approximately 1 MTHM
Commercial nuclear power plant and ISFSI sites storing UNF	73 sites
Sites with all nuclear power reactors permanently shut down	20 sites
Operating commercial nuclear power reactors	94 reactors
Permanently shut down commercial nuclear power reactors	27 reactors
DOE sites with UNF storage or research reactors	6 sites
University research reactor sites	20 sites
Other government research reactor sites	4 sites
Commercial research and development sites with UNF	4 sites
Primary reprocessing waste or high level waste sites	4 sites

^a The detailed DOE inventory total depends on the treatment of naval fuel and commercial fuel transferred to DOE custody. The value shown here follows the inventory basis used in this work. Quantities are rounded and may therefore differ slightly among inventory summaries.

The commercial inventory has also changed in composition over the history of the U.S. nuclear power

program. In the early decades of light water reactor operation, discharged fuel commonly had burnups below 20 GWd/MTM, and initial enrichments were often near 2 wt% ^{235}U . Improvements in fuel design, reactor operation, and fuel management strategies allowed utilities to extend cycle lengths and extract more energy from each assembly. By the twenty-first century, average discharge burnup had increased to approximately 50 GWd/MTM, while typical initial enrichment approached 4.5 wt% ^{235}U [2].

Figure 2.1 shows the increase in annual average burnup and initial enrichment for U.S. light water reactor discharges between 1968 and 2022. Although pressurized water reactor and boiling water reactor fuel follow somewhat different histories, both exhibit the same broad trend toward higher enrichment and higher discharge burnup.

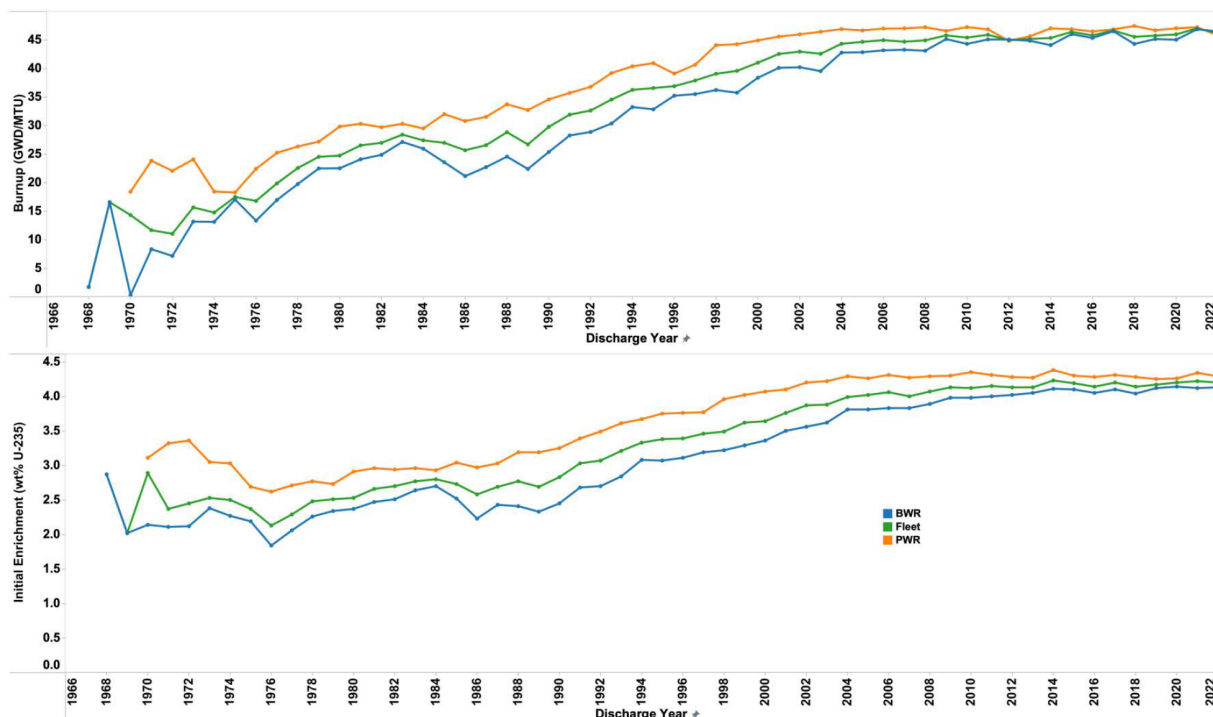


Figure 2.1: The annual average discharge burnup and initial ^{235}U enrichment of U.S. light water reactor fuel increased substantially between 1968 and 2022. Figure reproduced from [2].

Figure 2.2 provides the corresponding joint distribution of assembly burnup and initial enrichment in the inventory, showing both the central concentration of the data and the smaller number of assemblies at the extremes of the operating range.

Higher burnup changes the isotopic composition of discharged fuel. During irradiation, neutron capture in ^{238}U produces plutonium isotopes, while successive neutron capture and decay reactions generate neptunium, americium, and curium. Extending irradiation therefore generally increases the production of plutonium and minor actinides within an assembly, although the exact inventory depends on enrichment, reactor type, neutron spectrum, power history, and cooling time. The modern UNF inventory is consequently not uniform: assemblies discharged in different decades contain different quantities of fissile material, transuranic elements, fission products, and decay heat [2].

These compositional differences are important because the nuclides that control UNF behavior change with cooling time. Shortly after discharge, fission products dominate decay heat. In particular, ^{137}Cs and

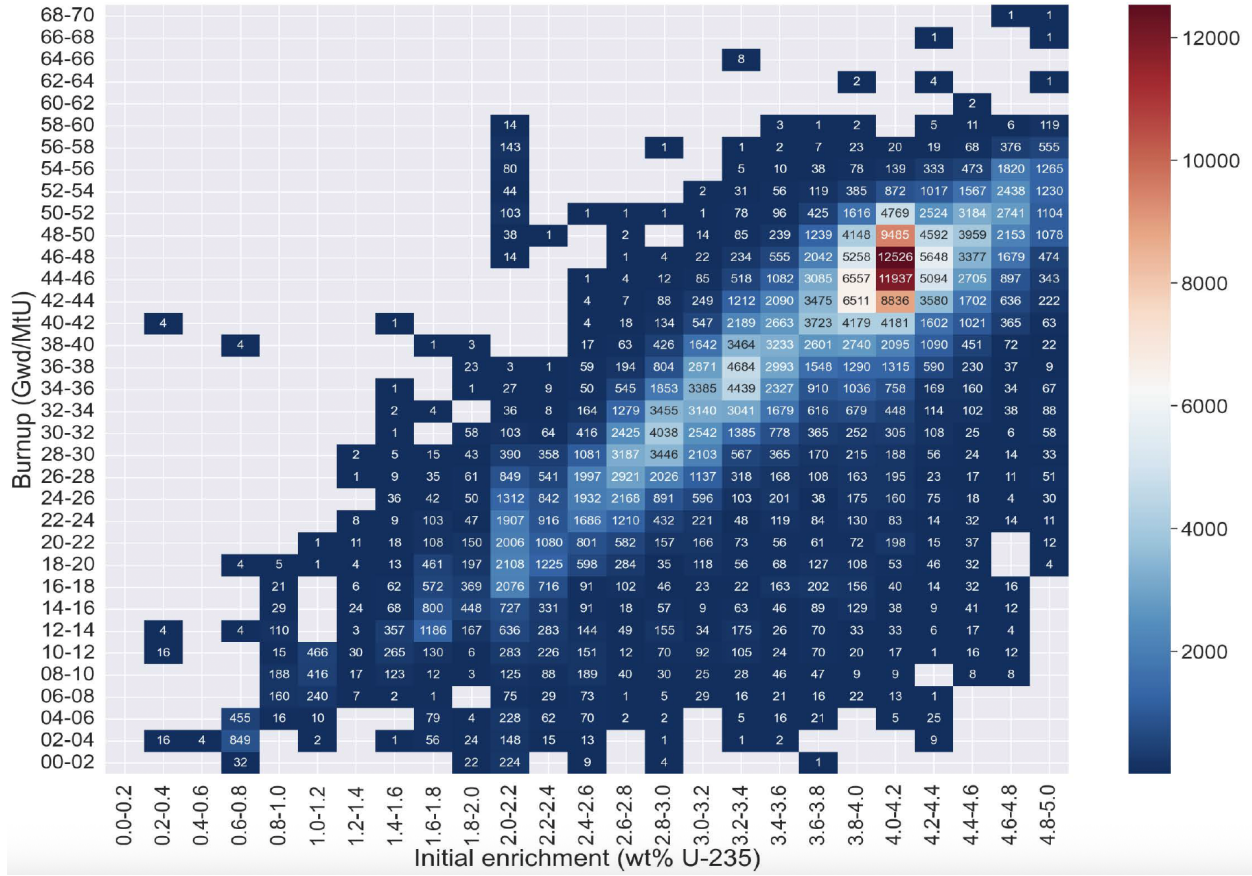


Figure 2.2: Burnup (GWd/MTHM) and initial enrichment (% ^{235}U) distributions for spent nuclear fuel assemblies in the Light Water Reactor (LWR) inventory through December 2022. The color scale and the value in each cell indicate the number of assemblies in the corresponding burnup–enrichment bin. Figure reproduced from [2].

^{90}Sr , together with their decay products, are major contributors over the first several decades. Their heat output decreases substantially as they decay. Over longer periods, transuranic isotopes, especially plutonium and americium isotopes, become the dominant contributors to the remaining heat and radiotoxicity [2].

This transition is significant for the U.S. waste management strategy. Storage can allow short-lived fission products to decay, but short-term storage does not eliminate the fundamental long-term burden posed by plutonium and the minor actinides. In a once-through fuel cycle, these nuclides must otherwise remain isolated while they decay over much longer timescales. Alternatively, actinides can be separated and irradiated in a suitable neutron spectrum, where fission converts them into shorter-lived fission products while releasing additional energy. Long-lived fission products may also be candidates for specialized transmutation, but the actinides are the principal component of UNF that can be destroyed through fission rather than managed solely by long-term storage and geologic isolation [16].

The increasing mass, geographic dispersion, and transuranic content of the U.S. UNF inventory therefore motivate the fuel cycle simulations in this work. A realistic assessment of accelerator-driven transmutation must represent not only the total mass of available fuel, but also the differences among assemblies in discharge date, burnup, enrichment, location, and isotopic composition. These attributes determine which portions of the inventory can be selected, separated, and supplied to an accelerator-driven system.

2.2 The CYCLUS Fuel Cycle Simulation Framework

The two archetypes developed in this thesis are implemented within CYCLUS, and their behavior can only be understood in relation to the framework’s underlying model of facilities, materials, and exchange. This section reviews the aspects of CYCLUS most relevant to that implementation and introduces the agent-based structure of the framework and how it represents fuel cycle facilities and their interactions over time. Then moving on to describe how materials and their isotopic compositions are represented and how radioactive decay is applied, which is central to tracking the evolving minor actinide vector on which this work depends and explains the dynamic resource exchange (DRE), the mechanism through which facilities request, offer, and trade resources at each time step.

2.2.1 Framework Basics

CYCLUS is an open-source, agent-based nuclear fuel cycle simulation framework designed to represent the deployment, operation, and interaction of nuclear fuel cycle facilities over time. The framework tracks discrete facilities and material resources rather than representing the fuel cycle solely through aggregated fleet-level flows. This design supports analyses in which individual facility behavior, material availability, and institutional decisions influence system level outcomes [10].

Agents in a CYCLUS simulation are organized into three hierarchical types: regions, institutions, and facilities. Regions form the highest organizational level and may represent countries, geographic areas, or other broad groupings. Institutions operate within regions and manage the deployment and retirement of facilities. Facilities represent physical or functional components of the nuclear fuel cycle, including mines, enrichment plants, reactors, separations facilities, storage facilities, and repositories [10].

A CYCLUS simulation advances through discrete time steps, commonly defined as months. During each time step, facilities may request resources, offer available resources, conduct material transactions, process

their inventories, and update their internal states. Modeling facilities and time discretely allows the framework to capture effects such as asynchronous reactor refueling, temporary shortages of fuel cycle materials, and differences in facility deployment and retirement schedules [17].

The behavior of each agent is defined by an *archetype*. Archetypes are modular software components that implement the physical behavior, operating logic, or decision rules of a region, institution, or facility. They are compiled independently from the CYCLUS kernel and loaded dynamically at the beginning of a simulation. This plug in architecture allows alternative models of the same fuel cycle process to be exchanged without modifying the core simulation framework [10,18].

The CYCLUS kernel provides shared simulation services, including time management, agent deployment, resource tracking, data recording, and resource exchange. Archetypes access these services through the agent application programming interface. As illustrated in Figure 2.3, the kernel provides the central simulation environment, the agent interface connects the kernel to agent models, and independently developed archetype libraries provide the behavior of the simulated regions, institutions, and facilities.

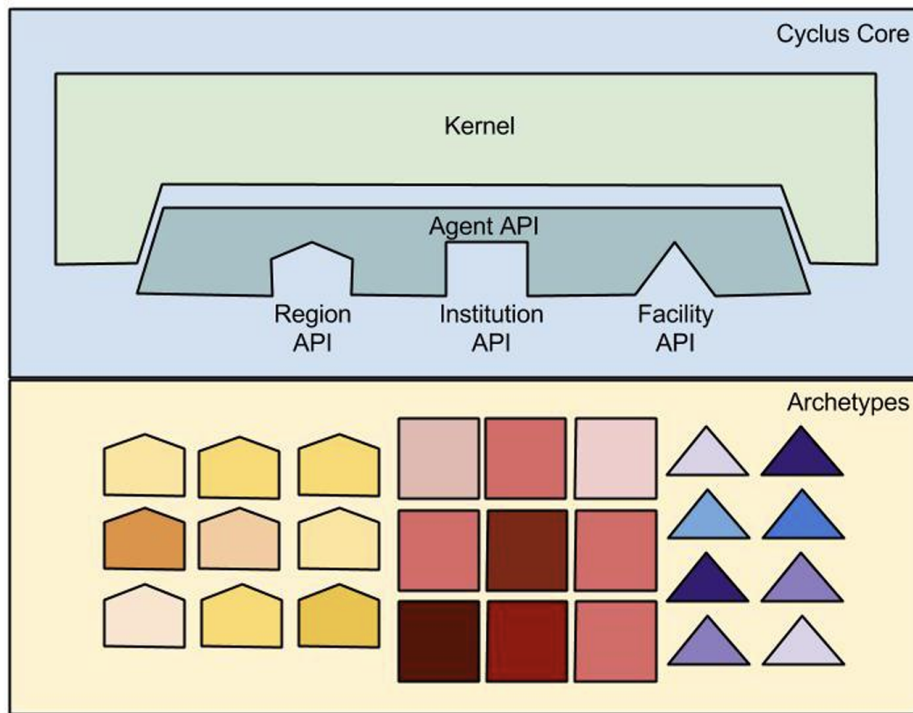


Figure 2.3: The CYCLUS simulation architecture separates the core kernel from dynamically loaded region, institution, and facility archetypes through the Agent application programming interface (API). Figure reproduced from [10].

2.2.2 Materials and Decay

Resources exchanged in CYCLUS are represented as discrete objects. The two primary resource types are *materials* and *products*. A material resource has an associated quantity and isotopic composition, whereas a product represents a non-nuclear resource whose quality may be described without a nuclide vector. Nuclear fuel cycle simulations therefore generally use material resources to represent fresh fuel, spent nuclear fuel, separated actinides, and radioactive waste [10].

A CYCLUS material object contains a total mass and a *Composition* object. The composition stores the relative abundance of the nuclides present in the material and is treated independently from the material quantity. This separation allows multiple material objects to reference the same composition and permits operations such as splitting, combining, and transmuting material while preserving mass accounting. Materials may also be stored in facility inventories and exchanged between agents through the resource exchange system.

Radioactive decay can be applied directly to material resources within CYCLUS. When a material is decayed over a specified time interval, its nuclide vector is updated according to the radioactive decay chains available to the framework. The decay calculation uses nuclear data supplied through PyNE [19], which is an open-source nuclear engineering toolkit that supplies the nuclear data including half-lives, decay constants, branching relationships, and daughter products. The total quantity of a material remains associated with the resource while its isotopic composition changes with time [10].

Decay in the simulations developed for this work is handled at the simulation level through the CYCLUS control block, with decay set to `lazy`. Under this setting the kernel decays a material’s composition on demand rather than at every time step, computing the decayed nuclide vector when a composition is next inspected and caching the result for reuse [10]. This choice retains the physical effect of radioactive decay over the multi-decade simulation horizon while avoiding the cost of decaying every inventory at every step. Decay is significant over this horizon: the roughly 14-year half-life of ^{241}Pu , for example, produces appreciable ingrowth of ^{241}Am across a thirty-year run, altering the minor actinide vector that the inventory and accelerator-driven system models are designed to track.

2.2.3 The Dynamic Resource Exchange

Transactions among CYCLUS facilities are coordinated through the DRE. The DRE provides a general mechanism through which independently developed archetypes can exchange resources without requiring direct knowledge of one another’s internal implementation. Facilities communicate their demand and supply to the framework, and the exchange system determines a feasible set of transactions for each time step [10].

The exchange begins when consuming facilities generate requests for resources. A request identifies a desired commodity, quantity, and, when relevant, a target material composition or other resource quality. Supplying facilities evaluate these requests and generate bids describing the resources they are able to provide. A bid links a particular offered resource to a request and may provide all or only part of the requested quantity [17].

Facilities may also assign preferences to possible trading partners or resources. Preferences allow a requester to rank otherwise feasible bids, for example by favoring a particular supplier, fuel composition, or material characteristic. In addition, suppliers and requesters may impose constraints such as production capacity, inventory availability, throughput, or minimum and maximum transaction quantities. These constraints define which combinations of trades are physically or operationally feasible.

The requests, bids, preferences, and constraints form a resource assignment problem. CYCLUS formulates this assignment as an optimization problem, solvable in either a linear or a mixed-integer form, and the DRE solves it to select a set of transactions that satisfies the applicable constraints while favoring higher preference trades [17]. After the solution is determined, the participating facilities execute the selected transactions and update their inventories. This process is repeated at each simulation time step, allowing supply and demand relationships to change dynamically as facilities are deployed, retired, depleted, or

otherwise alter their behavior.

By separating exchange coordination from facility specific behavior, the DRE allows archetypes to be developed and combined modularly. A storage facility, reactor, separations plant, or repository can participate in the same exchange system as long as it implements the appropriate request, bidding, and trade interfaces. This separation is particularly important for this work because the inventory and accelerator-driven system models must interact with both existing and newly developed fuel cycle facilities.

2.2.4 Extensions Developed in This Work

This work extends the CYCLUS ecosystem and its `cycamore` archetype library [20] with two facility archetypes developed within the `einstein` archetype library. The `us_inventory` facility represents the U.S. spent nuclear fuel inventory at assembly resolution, reading isotopic compositions and assembly metadata from the STANDARDS 5.0.1 dataset [8,9] and supplying material according to user-selected inventory selection policies. The `ads` facility represents an accelerator-driven subcritical burner, transforming a loaded core composition using a library of precomputed Serpent depletion cases. Both archetypes participate in the resource exchange described above: `us_inventory` as a supplier whose selection policies determine which assemblies it offers against a given request, and `ads` as a requester whose feed preferences rank inventory and separations output. Their design, implementation, and the four simulation scenarios in which they are exercised are described in Chapter 3.

2.3 Accelerator-Driven Systems

ADSs combine a high-power particle accelerator with a subcritical nuclear assembly. They have been proposed as dedicated systems for the transmutation of plutonium, minor actinides, and selected long-lived fission products recovered from spent nuclear fuel [16]. The ADS concept is particularly relevant to this work because it provides a potential pathway for consuming materials that is rich in minor actinides that present significant neutronic and safety challenges in critical reactors.

2.3.1 System Components and Operation

An ADS consists of three principal components: a high-energy proton accelerator, a spallation target, and a subcritical multiplying core. The accelerator produces a proton beam, often with an energy of several hundred megaelectronvolts to approximately one gigaelectronvolt. The beam strikes a heavy metal target, where high energy proton-nucleus interactions release neutrons through the spallation process. These source neutrons enter the surrounding subcritical core and induce fission reactions in the fuel [21,22]. The target must generate a sufficiently intense neutron source while withstanding substantial beam heating, radiation damage, and mechanical loading. Heavy elements are attractive target materials because they produce relatively large numbers of neutrons when bombarded by high-energy protons. Proposed ADS targets include solid tungsten targets and liquid metal targets based on lead or lead-bismuth eutectic. In some designs, a circulating liquid metal target also contributes to heat removal [21,22]. The core is designed to remain subcritical, such that

$$k_{\text{eff}} < 1, \tag{2.1}$$

where k_{eff} is the effective neutron multiplication factor. Unlike a critical reactor, the neutron population in an ADS core cannot sustain itself indefinitely without the external spallation source. Source neutrons initiate fission reactions and are multiplied within the subcritical core, allowing the thermal power produced by fission to exceed the power delivered by the proton beam [22,23].

The accelerator therefore provides an additional mechanism for controlling the fission power. When the beam is interrupted, the external source of spallation neutrons is removed and the source-driven fission power decreases rapidly. This behavior is sometimes described by stating that the beam acts as an on-off switch for the fission chain. However, turning off the beam does not eliminate radioactive decay heat. The fuel and structures continue to generate heat after shutdown and require reliable residual heat removal [23].

Figure 2.4 illustrates the relationship among the accelerator, spallation target, and subcritical core.

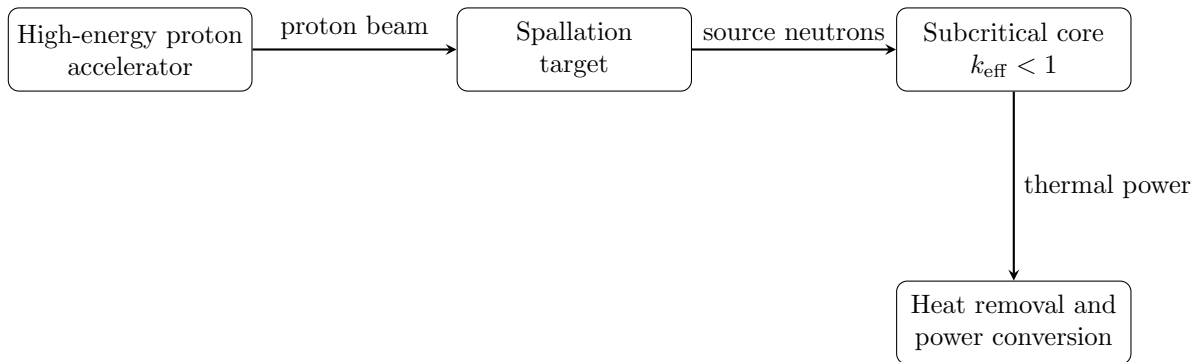


Figure 2.4: Principal components and energy flows of an accelerator-driven subcritical system. This original schematic is based on the ADS descriptions presented in [21,22].

Accelerator-driven transmutation of used nuclear fuel received sustained attention in the United States during the late 1990s and early 2000s, most prominently through the Accelerator Transmutation of Waste (ATW) program, which developed system scenarios and integration roadmaps for dedicated subcritical burners [24]. Interest in the approach has since reemerged in the context of managing the growing national inventory.

2.3.2 Minor Actinide Transmutation

The principal motivation for using ADS technology for nuclear waste management is its ability to accommodate fuels containing large fractions of minor actinides. The minor actinides most commonly considered for transmutation are neptunium, americium, and curium. These nuclides contribute to the long-term radiotoxicity and decay heat of spent nuclear fuel. Their separation and irradiation in a dedicated transmutation system could reduce the quantity of long-lived transuranic material requiring disposal [16].

Fast neutron spectra are generally preferred for minor actinide transmutation because fission becomes more competitive with radiative capture for many transuranic isotopes at higher neutron energies. A fast spectrum system can therefore destroy minor actinides through fission rather than merely converting them into heavier actinides [16].

However, loading a critical fast reactor with a high concentration of minor actinides introduces important safety and control challenges. Fuel rich in minor actinides generally has a smaller effective delayed neutron fraction than conventional uranium- or plutonium-bearing fuel. Although delayed neutrons represent only

a small fraction of the total fission neutron population, they substantially slow the response of a critical reactor to reactivity changes. A reduction in the effective delayed neutron fraction therefore causes the reactor power to respond more rapidly to disturbances [23].

Minor actinide loading may also reduce the magnitude of negative Doppler reactivity feedback. In conventional reactor fuel, increasing fuel temperature broadens neutron absorption resonances and generally introduces negative reactivity. This prompt feedback opposes the temperature increase. The magnitude of this effect can be weaker in minor actinide rich, fast spectrum fuels because of their isotopic composition and the reduced importance of resonance absorption in a fast spectrum [16,23].

The combination of a reduced delayed neutron fraction and weaker negative Doppler feedback makes a critical core with a high minor actinide content more difficult to control. Critical fast reactor concepts may therefore limit the amount of minor actinides loaded in the core or distribute them among fuel containing larger quantities of uranium or plutonium [23].

Subcritical operation provides an alternative approach. Because the core remains below criticality and relies on an external neutron source, an ADS can accommodate fuel compositions that would be difficult to use at high concentrations in a critical reactor. Subcriticality provides additional margin against prompt criticality and separates, to some extent, the control of the neutron source from the inherent reactivity of the fuel [23].

Subcriticality does not eliminate the need to evaluate reactivity accidents, cooling failures, material behavior, or decay heat removal. Nevertheless, it relaxes an important neutronic constraint on the amount of minor actinides that can be loaded into a dedicated transmutation system. This capability is the central reason that ADS technology is relevant to this thesis. The purpose of the modeled fuel cycle is to connect separated material from the U.S. spent nuclear fuel inventory with systems capable of consuming minor actinides streams that may not be suitable for concentrated use in critical reactors.

2.3.3 Target and Coolant Options

ADS concepts differ in their target materials, coolants, fuel forms, neutron spectra, and power levels. Lead and lead-bismuth eutectic are among the most frequently proposed heavy liquid metals because their large atomic masses support spallation neutron production and their high boiling temperatures permit low pressure primary systems [21,22].

Lead-bismuth eutectics can have a substantially lower melting temperature than pure lead, reducing some of the operational challenges associated with coolant freezing. The most advanced current example is Multi-purpose Hybrid Research Reactor for High-tech Applications (MYRRHA), a lead-bismuth-cooled subcritical system under development at the Belgian Nuclear Research Centre. Its first phase, the MYRRHA Injector and low-energy accelerator (MINERVA) proton accelerator and target facility, began construction in 2024 [25], making MYRRHA the closest existing demonstration of the accelerator driven concept. However, neutron activation of bismuth produces polonium-210, which creates radiological and maintenance concerns. Both lead and lead-bismuth can also cause corrosion or erosion of structural materials. Their use therefore requires careful control of coolant chemistry, oxygen concentration, temperature, and flow conditions [22].

Pure lead avoids the production of polonium associated with bismuth but has a higher melting temperature. This increases the importance of freeze prevention during startup, shutdown, and off-normal operation. Despite this challenge, lead has been proposed both as a coolant and as a carrier medium for mobile fuel ADS concepts [22,26].

The ADS concept that provides the initial technical basis for the EINSTEIN project was developed to transmute material derived from the U.S. spent nuclear fuel inventory. The proposed system uses a high energy proton accelerator, a spallation neutron source, and a lead-based subcritical system. The fuel concept contains plutonium and minor actinides in a mobile lead-based medium, permitting continuous or periodic processing of the fuel inventory [26].

Gas cooled ADS concepts have also been investigated. Helium is chemically inert and avoids many of the corrosion and activation issues associated with heavy liquid metals. However, its low density and volumetric heat capacity require high pressure operation and significant pumping power. The coolant and target selection therefore influences neutron production, material compatibility, plant efficiency, operating pressure, safety behavior, and the design of the primary heat removal system [22].

2.3.4 Technical and Economic Limitations

Despite their potential advantages for transmutation, ADS facilities face substantial technical and economic challenges. Accelerator availability is one of the most important engineering concerns. A sudden beam interruption causes a rapid decrease in fission power and temperature. Restoration of the beam causes the power and temperature to increase again. Repeated beam interruptions can therefore impose cyclic thermal loads on the fuel, spallation target, coolant systems, and structural components [27].

The Organisation for Economic Co-operation and Development (OECD) Nuclear Energy Agency beam interruption benchmark specifically examined the temperature transients caused by beam trips in a lead-bismuth-cooled, subcritical system. The study identified thermal fatigue of structural components as an important consequence of repeated beam interruptions [27]. More recent analysis of the China Initiative Accelerator Driven System has also evaluated fuel cladding fatigue under repeated beam trip transients [28].

Producing a continuous, high-current proton beam with the availability required for a power producing nuclear system is more demanding than operating an accelerator intended for intermittent research use. Fault-tolerant accelerator designs, redundant components, and rapid recovery procedures may reduce the number and duration of beam interruptions, but these features add complexity and cost [22].

The spallation target presents additional engineering challenges. It must tolerate concentrated beam heating, radiation damage, gas production, thermal stresses, and interactions with the coolant. Target replacement, inspection, and maintenance must also be performed in a highly radioactive environment [22].

ADS deployment additionally requires supporting fuel cycle infrastructure. Spent nuclear fuel must first be processed to recover the actinides intended for transmutation. The resulting material rich in minor actinides must then be fabricated into fuel or introduced into an appropriate mobile fuel medium. After irradiation, the material may require repeated separation and recycle to achieve substantial actinide destruction. These activities introduce material losses, secondary waste streams, safeguards requirements, worker exposure, and additional facilities [16].

An ADS also combines the capital cost and regulatory requirements of a nuclear reactor facility with those of a high-power accelerator. Furthermore, the accelerator consumes a portion of the electricity generated by the system. Comparative OECD/Nuclear Energy Agency (NEA) studies have therefore emphasized that the potential transmutation benefits of ADS technology must be evaluated alongside its fuel cycle requirements, technical complexity, research and development needs, and economic performance [23].

These limitations make a system level fuel cycle analysis necessary. The performance of an ADS cannot be judged solely from the rate of transmutation inside its core. A complete assessment must also consider

the availability and composition of feed material, separations capacity, facility deployment, accelerator and reactor throughput, recycle requirements, and the production of secondary waste streams. These interactions motivate the CYCLUS simulation models developed in this work.

2.4 The EINSTEIN Program

The Enhanced Integrated Nuclear Systems for Transmutation and Efficient Isolation of Nuclides (EINSTEIN) project is part of the ARPA-E Nuclear Energy Waste Transmutation Optimized Now (NEWTON) program. The NEWTON program was established to support technologies that enable the transmutation of used nuclear fuel and reduce the burden placed on permanent waste disposal facilities [4]. The EINSTEIN project is led by the University of Illinois Urbana-Champaign [29].

NEWTON addresses the transmutation problem through three complementary technical categories. Category A focuses on technologies for generating and accelerating particle beams capable of initiating transmutation reactions. Category B focuses on the design, fabrication, and processing of targets or fuel forms containing the nuclides to be transmuted. Category C integrates the technologies developed in Categories A and B through system level modeling, life cycle analysis, techno-economic analysis, and databases of materials and components [4,30]. UIUC is a Category C team and therefore does not develop a single accelerator or transmutation target in isolation. Instead, it provides a common framework for comparing candidate systems and identifying the technical and economic conditions under which they could contribute to management of the U.S. used nuclear fuel inventory.

Within the EINSTEIN project, the term XDS refers collectively to subcritical transmutation systems sustained by an external neutron source. Within this framing, ADSs use an accelerator and spallation target to produce the source neutrons, while fusion-driven systems (FDSs) use neutrons generated by a fusion source. The two classes may differ substantially in neutron energy, source geometry, target and coolant materials, reliability constraints, and technology readiness. Nevertheless, both couple an external neutron source to a subcritical multiplying region and can therefore be represented within a common fuel cycle and assessment framework. The initial EINSTEIN analyses include a high-energy, proton-driven, lead-cooled ADS and a 14.1 MeV fusion-neutron-driven, salt-cooled system [31].

The central objective of EINSTEIN is to develop a data-informed framework for rapidly assessing accelerator- or fusion-driven system (XDS) technologies according to both their transmutation performance and their financial implications. The project connects nuclear data, neutronics, materials compatibility, fuel cycle behavior, and economic performance rather than evaluating each component independently. The official project description identifies the development of a comprehensive life cycle analysis framework spanning the full transmutation process, from the composition of the incoming used fuel stream through irradiation, processing, and the resulting waste or product streams [29]. Experimental and computational activities are used to reduce uncertainties in neutron transport, material properties, and component performance.

The UIUC project team is organized around several interacting technical areas as shown in Figure 2.5. Nuclear data and diagnostics work identifies reactions whose uncertainties significantly affect neutron multiplication and transmutation predictions and develops experimental strategies to reduce those uncertainties. Neutronics analysis provides source term, criticality, depletion, and transmutation calculations for selected XDS concepts. Materials research evaluates corrosion, irradiation damage, thermophysical properties, and the chemical behavior of lead coolants and molten salts. Experimental work investigates the stability and solubility of representative actinide surrogates and the compatibility of structural materials

with the candidate coolant and fuel media [31].

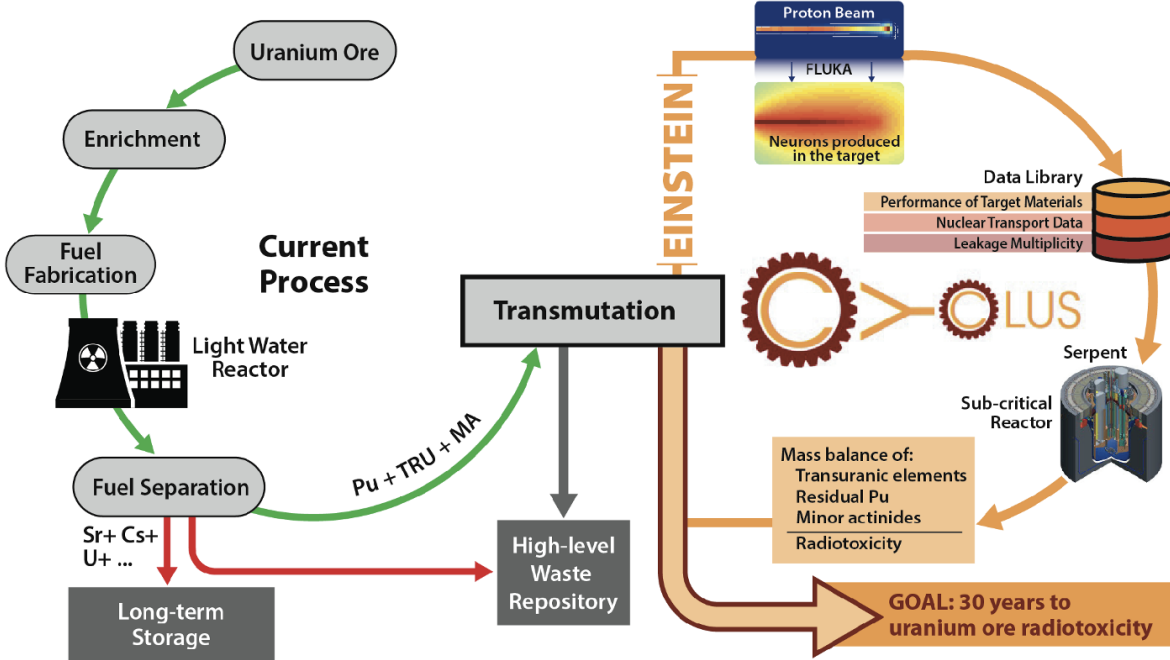


Figure 2.5: Overview of the UIUC EINSTEIN project, which integrates FLUKA, Serpent, nuclear data, materials, and CYCLUS fuel cycle analyses to evaluate accelerator-driven transmutation of plutonium and minor actinides from used nuclear fuel. The project aims to reduce the radiotoxicity of the remaining waste to approximately that of natural uranium ore within 30 years. Figure reproduced from [31].

These activities supply the inputs required for the fuel cycle and techno-economic assessment. The fuel cycle portion of the project represents the U.S. used nuclear fuel inventory, separations facilities, XDS facilities, recycle pathways, storage facilities, and waste streams in CYCLUS. The inventory characterization begins with national spent fuel records and age-corrected isotopic compositions, which are used to define realistic feed streams for the modeled XDS technologies. New CYCLUS archetypes are then used to evaluate how the choice of feed material, facility throughput, deployment schedule, separations capacity, and material selection policy affects system level transmutation performance [31].

The techno-economic analysis uses the outputs of the fuel cycle and component models to examine capital costs, operating costs, potential revenues, return on investment, and the sensitivity of economic conclusions to uncertain design parameters. This coupling is important because a design that performs well in a stand-alone neutronics calculation may still require an impractical number of facilities, excessive separations capacity, costly transportation, or unavailable supporting infrastructure. Conversely, a concept with a lower per-facility transmutation rate may be more attractive if it is easier to deploy, operate, and integrate with the existing nuclear fuel cycle.

The work presented in this thesis lies primarily within the fuel cycle modeling portion of EINSTEIN. It develops the `us_inventory` and `ads` archetypes in the `einstein` library and uses them to connect the characteristics of the national used fuel inventory with candidate XDS deployment scenarios. The resulting simulations provide material flow, facility scaling, and inventory depletion information that can be combined with the neutronics, materials, and economic analyses performed elsewhere in the project. In this

way, the thesis contributes the system level link between the physical design of an XDS and its potential effect on the U.S. used nuclear fuel inventory.

2.5 Prior Fuel Cycle Simulations of Transmutation Systems

Research on accelerator-driven transmutation has progressed through several distinct stages. Early studies established the physics of coupling an intense accelerator-produced neutron source to a subcritical multiplying medium. Subsequent programs developed and tested individual enabling technologies, including high current accelerators, liquid metal spallation targets, subcriticality monitoring methods, minor actinide fuels, and lead or lead-bismuth coolant systems. More recent studies have examined how dedicated transmutation facilities could be incorporated into regional or national nuclear fuel cycles.

These bodies of work do not all address the same modeling question. Reactor physics studies generally evaluate neutron multiplication, power distributions, safety coefficients, and nuclide destruction within one core. Technology demonstrators test particular components or operating procedures. Fuel cycle scenario studies instead examine the deployment of many reactors and supporting facilities over several decades, tracking quantities such as plutonium and minor actinide inventories, separations requirements, fuel fabrication capacity, and waste production. The present work belongs primarily to the third category, but its facility models are informed by the first two.

2.5.1 European Demonstrator Programs

Europe has sustained the most continuous multinational program of ADS research. Its development pathway has included conceptual system studies, critical and subcritical experiments, a megawatt class liquid metal spallation target experiment, and the staged development of the MYRRHA research facility. These programs have demonstrated important elements of an ADS, although no European facility has yet operated as a high power, minor actinide burning ADS.

One influential early European concept was Carlo Rubbia’s Energy Amplifier. The Energy Amplifier combined a high-energy proton accelerator, a heavy metal spallation target, and a subcritical fast spectrum blanket. Early versions emphasized energy production from thorium, while later analyses also considered waste transmutation and fast spectrum operation [32]. The importance of this work was not that it produced a directly deployable fuel cycle design, but that it articulated a complete accelerator target blanket system and stimulated coordinated European research on spallation targets, lead-based coolants, subcritical reactor physics, and partitioning and transmutation.

European research was subsequently coordinated through programs including EUROTRANS, which investigated an experimental ADS and the European Facility for Industrial Transmutation (EFIT). EFIT was conceived as a lead-cooled, fast spectrum minor actinide burner rather than as a general-purpose power reactor. Its design studies addressed accelerator requirements, target integration, subcritical core design, dedicated minor actinide fuel, and the supporting closed fuel cycle. In the “double-strata” strategy commonly associated with these studies, conventional critical reactors would manage most uranium and plutonium recycling, while a smaller fleet of dedicated ADS facilities would receive and destroy concentrated minor actinides [6].

The EFIT studies are particularly relevant to fuel cycle analysis because they treated an ADS not as an isolated core but as one component of a broader fleet. Regional scenario calculations examined the number of ADS facilities required to stabilize or reduce minor actinide inventories and estimated the associated

separations and fuel fabrication capacities. These analyses demonstrated that transmutation performance depends not only on the destruction rate of one ADS, but also on cooling delays, reprocessing losses, fuel fabrication throughput, deployment dates, and the availability of separated feed material [6].

The principal European engineering demonstrator is MYRRHA developed by Belgian Nuclear Research Centre (SCK CEN) in Belgium. MYRRHA is designed as a lead-bismuth eutectic cooled, fast spectrum research system capable of subcritical accelerator driven operation and, in the broader design concept, critical operation. Its purposes include demonstrating ADS technology, conducting fuels and materials irradiations, supporting radioisotope production, and providing experimental data relevant to minor actinide transmutation [21,25].

MYRRHA is being implemented in phases. The first phase, MINERVA, comprises the lower-energy portion of the proton accelerator and associated target stations. Construction of this phase began in 2024. The later phases would extend the accelerator and couple it to the subcritical reactor. MYRRHA therefore represents the most advanced European program directed toward a power scalable ADS, but it should not yet be described as an operating accelerator-driven reactor [25].

Several predecessor experiments were developed to reduce the technical risk of MYRRHA. The Generator of Uninterrupted Intense Neutrons at the Lead VENUS Reactor (GUINEVERE) project coupled the Générateur de Neutrons Pulsés Intenses, third generation with continuous beam (GENEPI-3C) deuteron accelerator to the fast-spectrum configuration of the Vulcan Experimental Nuclear Study reactor (VENUS-F) fast lead critical and subcritical assembly at SCK CEN. GUINEVERE was designed principally to investigate reactor physics questions, including the determination and online monitoring of subcriticality, source jerk and beam interruption methods, core loading procedures, and the transition between critical and subcritical configurations [33–35]. It was a low-power experimental facility rather than a fuel burning demonstrator, but it provided evidence that an accelerator could be coupled to a fast subcritical assembly and that the system’s reactivity could be measured using realistic source operation modes.

A separate enabling technology experiment was MEGAwatt Pilot Experiment (MEGAPIE), operated at Swiss Spallation Neutron Source (SINQ) in 2006. MEGAPIE exposed a circulating lead-bismuth target to a continuous 590 MeV proton beam at approximately 0.77 MW. It was the first liquid metal spallation target operated in the megawatt beam regime and supplied experimental data on neutron production, heat removal, liquid metal operation, activation products, structural behavior, and post-irradiation examination [36–38]. MEGAPIE did not include a multiplying reactor core, so it did not demonstrate an integrated ADS. Its importance was instead the validation of one of the most demanding ADS components under an irradiation environment approaching that required for a full-scale system.

France contributed to these European programs through Commissariat à l’Énergie Atomique et aux Énergies Alternatives (CEA), National Centre for Scientific Research (CNRS), and other institutions, as well as through national fuel cycle scenario studies. CEA developed the Commelini-Sicard (COSI) simulation code to model reactor fleets and associated fuel cycle facilities over long time horizons. COSI studies examined transitions from light water reactors to sodium cooled fast reactors and compared homogeneous, heterogeneous, and dedicated minor actinide transmutation strategies [7]. Other French and European studies compared ADS-based transmutation with minor actinide burning in critical fast reactors, including differences in fuel cycle cost, inventory reduction, and supporting facility requirements [39,40].

Taken together, the European programs established a progression from component experiments to integrated demonstrator design and fleet-level scenario analysis. GUINEVERE addressed source coupling and subcriticality monitoring, MEGAPIE addressed a high power liquid metal target, EFIT provided a

reference industrial transmuter design, and MYRRHA seeks to integrate many of these capabilities into a research facility. Their scenario studies also showed that the viability of ADS deployment cannot be inferred from single core transmutation efficiency alone.

2.5.2 United States Transmutation Programs, 1990s–2000s

Modern U.S. accelerator-driven transmutation work grew partly from studies at Los Alamos National Laboratory in the late 1980s and early 1990s. Bowman et al. proposed coupling an intense accelerator-produced neutron source to a subcritical multiplying and transmuting medium. The concept considered the destruction of transuranic elements and selected long-lived fission products, together with energy production from fertile material [41]. This work helped establish the technical foundation for the (ATW) program.

ATW was developed during the 1990s as a system for treating commercial spent nuclear fuel using partitioning, accelerator-driven transmutation, and repeated recycle. A reference ATW complex included front-end separations, minor actinide and plutonium fuel preparation, one or more accelerator-driven subcritical burners, recycle separations, and waste conditioning. The U.S. Department of Energy submitted a development roadmap to Congress in 1999, identifying required research in accelerators, spallation targets, separations, fuels, materials, reactor physics, safety, and systems analysis [24,42].

The roadmap treated ATW explicitly as a fuel cycle system rather than merely as a reactor. This was important because repeated recycle is required to achieve high actinide destruction. No irradiation cycle destroys every atom fed to the system: some actinides remain in the discharged fuel, some are converted into other actinides, and small fractions are lost during chemical processing and fabrication. Consequently, the overall reduction in waste burden depends on the efficiency and capacity of the complete recycle loop. The follow-on Advanced Accelerator Applications (AAA) program broadened the scope of the U.S. effort to include waste transmutation, energy applications, and enabling accelerator technologies. Program planning documents considered the reliability of high power proton accelerators, liquid metal and solid targets, subcritical blanket configurations, separations based on aqueous and pyrochemical methods, and fabrication of fuels containing high fractions of transuranic elements. Much of this work remained at the level of design studies, laboratory research, and component development. No integrated U.S. ATW or AAA engineering demonstrator was constructed.

Systems studies performed for ATW and related programs compared combinations of light water reactors, critical fast reactors, and accelerator-driven systems. These studies investigated questions such as whether plutonium should first be recycled in light water reactors, whether ADS facilities should consume all transuranics or only minor actinides, and how many recycle passes would be required. Their results repeatedly showed that conclusions were sensitive to separations efficiency, fuel cooling time, accelerator availability, support ratio, and the assumed growth or decline of nuclear power [24,42].

After the large ATW and AAA programs ended, U.S. ADS work continued through more focused reactor physics, materials, target, and systems studies. A recent and particularly relevant example is the Argonne National Laboratory concept reported by Gohar, Cao, and Kraus. Their design was developed specifically to consume plutonium and minor actinides derived from the U.S. spent nuclear fuel inventory. It employs a 1 GeV proton beam, a lead-based spallation target, and a subcritical fast blanket with mobile fuel containing plutonium and minor actinides [26].

Gohar et al. used Monte Carlo transport and depletion calculations to examine several blanket compositions and estimate the actinide consumption achievable over decades of full-power operation. The study

concluded that a small fleet of large ADS units could consume the minor actinide inventory assumed in its analysis. However, its scope was a set of reference ADS designs and their depletion performance. It did not dynamically model the level U.S. inventory, site to site material movement, separations deployment, facility construction schedules, or competition among facilities for particular spent fuel resources. Thus, it provides the principal reactor design basis for the ADS model in this thesis, but not the system level model needed to determine how that design would be supplied and deployed.

2.5.3 Ongoing Construction: CiADS

China began a national ADS development program through the Chinese Academy of Sciences in 2011. Early work included the China Lead-based Reactor (CLEAR) program and a series of lead-bismuth experimental loops, zero-power facilities, and non-nuclear test systems. These activities supported the development of coolant technology, structural materials, fuel assemblies, components, and operating methods for a lead-cooled subcritical system [43].

The China initiative Accelerator Driven System (CiADS), led by the Institute of Modern Physics of the Chinese Academy of Sciences, is now under construction in Huizhou. The integrated facility is designed to couple a superconducting proton linear accelerator, a lead-bismuth spallation target, and a lead-bismuth-cooled subcritical reactor. Published design studies describe a first-stage reactor with a maximum thermal power of approximately 10 MW. The initial core uses oxide fuel and is intended primarily to demonstrate the engineering feasibility, control, safety, and coupled operation of the ADS rather than to serve immediately as an industrial-scale minor actinide burner [44,45].

CiADS is significant because it moves beyond isolated component tests toward construction of an integrated accelerator target subcritical reactor facility. Its research program addresses accelerator reliability, beam transport, target behavior, lead-bismuth technology, reactor control, thermal hydraulics, and protection systems. In the proposed operating strategy, core power is controlled through the proton beam power because the subcritical core does not contain a conventional critical reactor control system for routine power regulation [44,46].

Dynamic and transient studies of CiADS have analyzed beam trips, beam overpower, loss of flow, loss of heat sink, and station blackout conditions. These calculations indicate substantial thermal margins for the low power demonstrator while identifying longer term concerns such as lead-bismuth corrosion, creep, and fatigue under repeated thermal cycling [45]. The project therefore provides an important test of whether the theoretically attractive subcritical characteristics of ADSs can be realized in a maintainable engineering system.

CiADS should not be interpreted as a completed demonstration of a commercial transmutation fuel cycle. The first stage system is much smaller than the gigawatt thermal designs commonly assumed in industrial fuel cycle studies, and its initial mission emphasizes technology demonstration. Nevertheless, operational experience from CiADS could reduce major uncertainties presently treated parametrically in systems models, including beam availability, component lifetime, maintenance frequency, achievable capacity factor, and the coupling between accelerator and reactor operation.

2.5.4 Fuel Cycle Simulation Studies

The studies described above span a wide range of spatial and temporal resolutions. Core design calculations may represent individual fuel pins and detailed nuclide vectors but usually consider only one reactor

and a limited number of irradiation cycles. Fuel cycle simulators generally model decades of fleet evolution, but they aggregate fuel into batches, reactor types, or material streams. This trade-off between reactor detail and system breadth is central to understanding the gap addressed by this thesis.

The OECD/NEA regional scenarios study extended this type of analysis to groups of countries with different nuclear policies and fuel cycle infrastructure. It used the EFIT design as the reference ADS and calculated the number of transmuters and associated fuel cycle facilities required to stabilize minor actinide inventories across a regional system [6]. This work showed that regional sharing of specialized facilities could reduce duplication of expensive separations and fabrication plants. However, its material representation was at the level of national or regional inventory streams rather than individual discharged assemblies.

France’s COSI code has been used extensively for dynamic scenario analysis. COSI represents reactor fleets, fuel fabrication, irradiation, cooling, reprocessing, storage, and waste production over time. Applications have included transitions from the existing French light water reactor fleet to sodium-cooled fast reactors, along with comparisons of minor actinide management options [7]. These studies are highly detailed in their treatment of fleet evolution and industrial fuel cycle capacity, but they generally use reactor and fuel batches characterized by representative compositions rather than a geospatially explicit assembly inventory.

European projects also compared the economic and material consequences of minor actinide transmutation in critical fast reactors and ADS facilities. Chabert et al. compared no-transmutation, fast reactor, and ADS options, while Merino-Rodríguez et al. cross-checked economic and mass balance calculations for scenarios involving dedicated ADS burners [39,40]. These studies demonstrated that ADS-based strategies may achieve concentrated minor actinide destruction but incur additional costs associated with the accelerator, dedicated fuel fabrication, and supporting recycle facilities.

The Belgian Analysis of Nuclear-fuel-cycles Including Chain and Cost Assessment (ANICCA) code provides a more recent example directly related to ADS deployment. ANICCA is a dynamic nuclear fuel cycle simulator developed to study Belgian fleet evolution and waste management. A published verification and case study examined the effect of adding an ADS to a Belgian nuclear phase-out scenario. The model tracked fleet-level materials and showed the potential reduction of minor actinide inventories, but it did not model individual commercial fuel assemblies or source site specific selection [5].

Other studies have focused on detailed depletion within an ADS rather than deployment of the surrounding fuel cycle. These calculations optimize core geometry, fuel loading, neutron spectrum, or recycle strategy and report nuclide-specific transmutation rates. They are essential for supplying performance parameters to a system model, but they do not themselves answer how quickly suitable material becomes available, where it is stored, which assemblies are selected, or how infrastructure limits deployment [26].

Across the literature reviewed here, system level ADS studies generally use one of three material representations. The first is an equilibrium mass flow model, in which annual production and destruction rates are balanced without explicitly simulating each facility. The second is a dynamic fleet model in which reactor types and fuel cycle plants are deployed over time and exchange aggregated batches. The third is a detailed core or depletion model in which a single ADS is simulated at high isotopic resolution but the external fuel cycle is prescribed rather than dynamically represented.

These approaches have provided estimates of ADS support ratios, required fleet size, equilibrium actinide inventories, separations capacity, fuel cycle cost, and repository benefits. They have not generally represented the existing U.S. used fuel inventory as a set of discrete assemblies carrying site, discharge date,

burnup, enrichment, mass, and composition metadata. In the literature reviewed for this thesis, no prior ADS fuel cycle study was identified that dynamically selected individual assemblies from the site-resolved U.S. inventory and supplied them through separations to parameterized ADS facilities.

Table 2.2 summarizes the different resolutions and technical scopes represented by the major programs and simulation studies discussed above.

Table 2.2: Prior accelerator-driven transmutation programs and studies addressed different levels of technology and fuel cycle resolution. Information summarized from [5,6,25,26,33,37,45].

Program or study	Type	Principal scope	Material resolution
GUINEVERE	Experimental facility	Accelerator coupling, fast subcritical physics, and online reactivity monitoring	Core configurations; no fuel cycle deployment
MEGAPIE	Component experiment	Megawatt-class liquid lead-bismuth spallation target	Target and neutron source behavior only
MYRRHA	Staged demonstrator	Accelerator, target, fast spectrum research reactor, fuels, materials, and ADS operation	Core and irradiation experiments; supporting scenario studies
EFIT / European scenarios	Reference design and fleet studies	Dedicated minor actinide burning and regional deployment	Aggregated fuel batches and regional inventories
ATW / AAA	U.S. program and system studies	Separations, recycle, accelerator, target, and subcritical burner	Process streams and representative fuels
Gohar et al.	ADS design and depletion study	Lead-based mobile fuel ADS for the U.S. actinide inventory	Prescribed blanket compositions
CiADS	Integrated demonstrator under construction	Coupled accelerator, spallation target, and low-power subcritical reactor	Detailed core and component models
COSI / AN-ICCA studies	Dynamic fuel cycle simulation	Fleet transitions, facility capacities, inventories, and waste	Reactor fleets and aggregated fuel batches
This work	Dynamic fuel cycle simulation	U.S. inventory selection, separations, ADS deployment, throughput, and inventory evolution	Individual assemblies with associated metadata

That distinction is important for the U.S. case. The national inventory is not an equilibrium material stream produced by a uniform reactor fleet. It is a heterogeneous legacy inventory accumulated over several decades and stored at geographically dispersed operating and shutdown sites. Assemblies differ in age, burnup, enrichment, reactor type, remaining heavy metal mass, isotopic composition, and storage location. A model that begins with a representative average composition cannot evaluate how alternative inventory selection policies change the material supplied to separations and transmutation facilities.

The present work addresses this gap by combining an assembly resolved U.S. inventory facility with dynamically deployed ADS and supporting fuel cycle facilities in CYCLUS. It does not replace detailed neutronics, depletion, materials, or economic models. Instead, it provides the system layer through which those models can be connected to the actual characteristics and availability of U.S. used nuclear fuel. This representation permits analysis of inventory selection policies, throughput limits, facility scaling, deployment timing, material availability, and the evolution of the remaining national inventory.

2.6 Research Gap

The literature reviewed in Section 2.5 demonstrates substantial progress in accelerator-driven transmutation, but the principal areas of progress have remained separated. Reactor physics and engineering studies have developed increasingly detailed representations of accelerators, spallation targets, subcritical cores, fuels, coolants, and transmutation performance. Fuel cycle studies have examined the deployment of reactors, separations plants, fabrication facilities, and waste management infrastructure over long periods. National spent fuel databases have also developed sufficiently detailed records to characterize individual discharged fuel assemblies. However, these capabilities have not generally been combined in a single simulation. In the literature reviewed for this thesis, no published study was identified that dynamically couples a real, assembly resolved U.S. spent nuclear fuel inventory to an ADS facility inside a fuel cycle simulator and evaluates how the selection of particular assemblies affects transmutation system deployment and the evolution of the remaining national inventory.

European ADS research has produced the most mature combination of experimental facilities, component demonstrations, core calculations, and regional fuel cycle studies. GUINEVERE investigated accelerator coupling and subcriticality monitoring, MEGAPIE demonstrated a liquid metal spallation target under a megawatt-class proton beam, and MYRRHA has advanced the engineering design and staged implementation of an integrated ADS research facility [21,25,33,34,37]. European fuel cycle tools have also examined fleet transitions and minor actinide transmutation. For example, COSI has been used to evaluate French reactor and fuel cycle transitions, while ANICCA has modeled the introduction of an ADS into a Belgian nuclear phase-out scenario [5,7]. These studies provide detailed information about European reactor fleets, fuel cycle infrastructure, and transmutation strategies, but their feed materials are represented as reactor specific or scenario specific fuel streams and batches. They do not begin with the discrete assembly records of the existing U.S. inventory or evaluate alternative rules for selecting particular U.S. assemblies for processing.

The U.S. Accelerator Transmutation of Waste and Advanced Accelerator Applications programs addressed many policy relevant aspects of a complete transmutation system. Their roadmaps and system studies considered separations, repeated actinide recycle, fuel fabrication, accelerator reliability, spallation target design, subcritical burners, and final waste management [24,42]. These efforts recognized that the performance of an ADS could not be assessed independently from the facilities that prepare, recycle, and manage its fuel. Nevertheless, the principal ATW and AAA studies were conducted before the development of current agent-based fuel cycle simulators and before the availability of the present assembly level STANDARDS database. Consequently, their national inventory was represented through assumed aggregate material streams, reference isotopic compositions, and projected production rates rather than through the individual assemblies that constitute the existing U.S. inventory.

More recent U.S. work has improved the physical basis of the ADS itself. Gohar, Cao, and Kraus devel-

oped a lead-based ADS concept specifically for consuming plutonium and minor actinides derived from U.S. spent nuclear fuel and used detailed transport and depletion calculations to estimate actinide consumption over the operating life of the system [26]. This study provides the initial ADS design basis used in the present work, but its treatment of the national inventory begins with the total assumed masses of plutonium and minor actinides. It does not model the discharged assemblies from which those materials would be recovered, the order in which assemblies would be selected, or the time-dependent interaction among inventory, separations, and ADS facilities.

CiADS addresses a different part of the problem. Its published analyses focus on the engineering behavior of an integrated accelerator, spallation target, and low-power subcritical reactor. Studies have examined proton beam control, reactor transients, beam trips, thermal margins, coolant behavior, and structural fatigue [28,43–45]. This work is essential for demonstrating whether ADS technology can be operated reliably, but CiADS is an engineering demonstrator rather than a simulation of national fuel cycle deployment. Its analyses therefore do not address how a heterogeneous legacy inventory would be selected, transported, separated, and allocated among multiple future transmutation facilities.

Existing fuel cycle and inventory studies come closest to connecting these areas. The STANDARDS 5.0.1 Database contains detailed information about the spent nuclear fuel (SNF) inventory in the U.S. with full isotopic composition. Peterson et al. described the planned use of Unified Database (UDB) information in system models such as Next-Generation System Analysis Model (NGSAM) and in fuel cycle transition analyses using ORION [9]. Bae et al. subsequently trained a neural network on UDB depletion data and implemented the resulting model in CYCLUS, enabling efficient prediction of spent fuel compositions for reactor batches with varying burnup and enrichment [12]. These studies established that realistic U.S. data can improve fuel cycle simulation. The former described broader integration of the database with system and fuel cycle tools, while the latter reproduced UDB-derived isotopic behavior without exposing or dynamically transacting the underlying private assembly records. Neither work connected an active assembly inventory to an ADS burner or examined how the choice among available assemblies changes the feed supplied to that burner.

The gap addressed in this thesis is therefore narrower than the absence of either detailed ADS models or detailed inventory data. The missing capability is their integration at the system level. This work represents individual U.S. spent fuel assemblies as selectable resources within an agent-based fuel cycle simulation. Assembly records derived from STANDARDS 5.0.1 retain attributes relevant to material selection, including available heavy metal mass, discharge date, burnup, initial enrichment, and isotopic composition [2]. The `us_inventory` archetype supplies these resources through the CYCLUS Dynamic Resource Exchange rather than first collapsing the national inventory into a single average recipe.

The inventory model is coupled to an ADS archetype whose material transformation is based on precomputed Serpent depletion results. The simulation can therefore evaluate multiple inventory selection policies, including selection by availability, age, burnup, and enrichment, while maintaining material accounting as assemblies are supplied and their available masses are depleted. The results presented in Chapters 3–4 quantify how these policies change the mass and characteristics of material supplied to the modeled fuel cycle facilities and how they alter the composition of the inventory remaining after transmutation.

This integration does not by itself constitute a complete recommendation for U.S. waste policy. Questions involving facility siting, transportation approval, safeguards, separations readiness, cost, financing, and repository performance require additional models and institutional assumptions. Rather, the contribution of this thesis is to provide a U.S.-specific material flow foundation on which those analyses can be built.

By connecting an assembly resolved legacy inventory to an ADS representation in CYCLUS, the work enables transmutation scenarios to be evaluated using the actual heterogeneity and availability of U.S. spent nuclear fuel instead of an assumed national average feed.

Chapter 3

Methodology

This chapter describes the methods developed to connect a real, assembly-resolved U.S. spent fuel inventory to an accelerator-driven burner within a single fuel cycle simulation. The work required building two new facility archetypes for CYCLUS, `us_inventory` and `ads`, and assembling them together with existing fuel cycle facilities into complete transmutation scenarios. The chapter sets out the overall modeling approach, the language and software framework in which the archetypes are implemented, the design and data behind each archetype in turn, and finally the specific fuel cycle configurations and simulation runs used to generate the results presented in Chapter 4.

3.1 Overview of the Simulation Approach

The simulations in this work are performed with CYCLUS, the agent-based nuclear fuel cycle simulator introduced in Chapter 2. Two facility archetypes were developed within the `einstein` archetype library: `us_inventory`, which represents the United States inventory of UNF at the resolution of individual assemblies, and `ads`, which represents an accelerator-driven subcritical burner. These are combined with facilities from the CYCAMORE library, `Separations` to partition discharged fuel into product and waste streams, `Mixer` to blend separated streams into a burner feed, and `Sink` to receive final waste, to form a complete transmutation fuel cycle. The two variants of this cycle used in this work, once-through and recycling, are shown in Figures 3.4 and 3.5 in Section 3.5.1.

The remainder of this chapter describes each archetype in turn, `us_inventory` in Section 3.3 and `ads` in Section 3.4, and then defines the specific scenarios built from them in Section 3.5.

All scenarios share a common set of simulation settings. Time advances in monthly steps of 30.4375 d, beginning in January 2025, over a horizon of either 30 years (360 steps) or 100 years (1200 steps). Radioactive decay is enabled at the simulation level through the CYCLUS control block with decay set to `lazy`, so that material compositions are decayed on demand over the multi-decade horizon rather than at every time step [10]. The full set of runs and the parameters for each scenario are given in Section 3.5.

3.2 Implementation within Cyclus

Both archetypes are written in C++, the language of the CYCLUS kernel itself [10]. Each is implemented as a header and implementation file pair within the `einstein` archetype library, under a shared `einstein`

namespace, following the same structure and conventions as the standard CYCAMORE library on which it is modeled [20]. The library is maintained as an open-source repository on GitHub¹, alongside the CYCLUS kernel², and the CYCAMORE archetypes³, and is compiled into a dynamically loadable archetype library that the CYCLUS kernel loads at runtime according to the archetypes named in a simulation’s input file.

An archetype becomes a CYCLUS facility by inheriting from the framework’s base classes rather than by reimplementing simulation infrastructure. The `us_inventory` archetype derives from `cyclus::Facility`, which in turn derives from `cyclus::Agent`, and additionally from `cyclus::toolkit::CommodityProducer` so that it is recognized as a producer of a configured output commodity; the `ads` archetype derives from `cyclus::Facility` alone. A single header, `cyclus.h`, exposes the kernel interface used throughout: the simulation `Context`, the `Material` and `Composition` resource types, the resource exchange objects through which facilities bid on and trade material, and the toolkit utilities on which the archetypes rely. Inheriting from these classes places each archetype directly into the agent-based scheduling and DRE machinery described in Chapter 2, so that development can focus on facility behavior rather than on the mechanics of the simulation loop.

The remaining connective code is generated automatically by the CYCLUS preprocessor, `cycpp`, which processes `#pragma cyclus` annotations embedded in the source. Persistent state is declared by marking a member variable with `#pragma cyclus var`, which records the information the kernel needs to validate that variable against the input schema, write it to and restore it from the simulation database, and copy it when an agent is cloned. Companion directives request the default generated implementations of these operations and attach human-readable documentation to the archetype. This arrangement keeps the input specification, database schema, and internal state of each archetype consistent without handwritten boilerplate, so that the source files contain primarily the physical and logical behavior of the inventory and burner models that the remainder of this chapter describes.

3.3 The `us_inventory` Archetype

The `us_inventory` archetype represents the existing U.S. UNF inventory as an active material supplier. Rather than collapsing the national inventory into a single resource with a prescribed mass and representative composition, it preserves individual assembly records, because the U.S. inventory is heterogeneous and the order in which assemblies are selected changes the mass and isotopic composition delivered to downstream facilities.

Each assembly is stored as a separate inventory assembly with its own mass, operating history, storage information, composition, and geographic provenance. During a simulation the facility answers material requests through the CYCLUS DRE (Chapter 2) and selects an eligible assembly according to a user-specified policy, reducing that assembly’s mass and the total inventory as material is supplied. It inherits from `cyclus::Facility` and `cyclus::toolkit::CommodityProducer`, so it acts as both a facility agent and a producer of a configured output commodity. The facility performs no scheduled action each time step; all of its behavior is triggered either when the input data are loaded at initialization or when it participates in resource exchange. The overall structure of the archetype is illustrated in Figure 3.1.

¹<https://github.com/arfc/einstein>

²<https://github.com/cyclus/cyclus>

³<https://github.com/cyclus/cycamore>

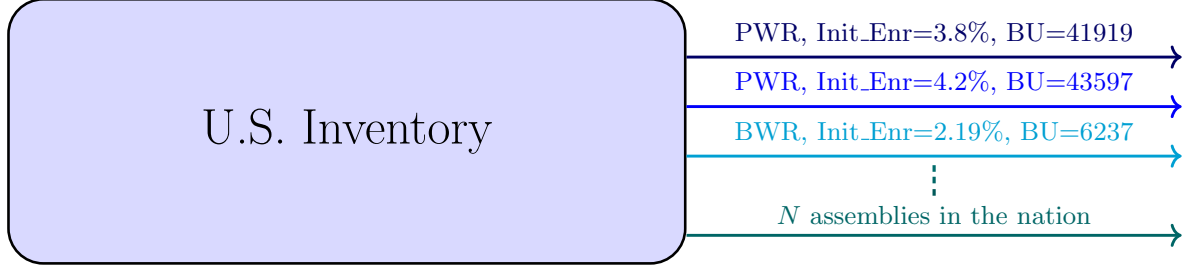


Figure 3.1: Structure of the U.S. Inventory module based on the STANDARDS 5.0.1 dataset. Representative PWR and BWR assemblies are identified by their initial enrichment, burnup, and isotopic composition, and the module retains all assemblies in the national spent fuel inventory.

3.3.1 Purpose and Assembly-Level Data Source

The source data are assembly-level records derived from STANDARDS 5.0.1, which associates individual U.S. spent fuel assemblies with facility, storage, irradiation history, and calculated isotopic information. This level of detail distinguishes assemblies that would otherwise be represented using an average composition for a facility or for the national inventory [9,47].

For the simulations in this thesis, the STANDARDS data were provided as a directory of 75 line-delimited files, one per represented facility. The file count is not hard-coded: the `data_dir` parameter names a directory, and every file whose name ends in `.txt` or contains `.jsonl` is read, so the archetype can run on a subset of the database without recompilation.

Each nonempty line is one assembly record, converted into assembly metadata and a nuclide mass map. Because the export wraps each dictionary-like record in an outer JSON string, a purpose-built parser removes that wrapper and accepts both single- and double-quoted strings. It reads the facility name, storage location name, assembly identifier, storage type, discharge year, initial enrichment, maximum burnup, initial uranium mass, and a dictionary of nuclide masses in grams. Unrecognized fields are ignored, and records with no nonzero composition are skipped.

The nuclide names are converted to the integer ZZAAAM identifiers used by CYCLUS and assembled into a `cyclus::Composition`; the total assembly mass is converted from grams to kilograms and stored with the record. Each assembly is held in an `assembly` structure, summarized in Table 3.1.

Table 3.1: Fields stored in each `assembly` structure.

Field	Description
Assembly_ID	Original database assembly identifier.
Facility, location	Source facility and storage location names.
Storage type	Wet or dry storage.
Available mass	Current material mass, in kilograms.
Discharge year	Year the assembly was discharged.
Maximum burnup	Reported maximum burnup.
Initial enrichment	Initial uranium enrichment.
Initial uranium mass	Initial mass of uranium.
Derived fractions	Fissile, plutonium, and minor actinide mass fractions.
Coordinates	Latitude and longitude of the source facility.
Composition	Complete CYCLUS composition of the assembly.

Because the original identifier is not unique across the dataset, the archetype builds an internal key from

the facility name, storage location, original identifier, and load position, stored in `assembly_id`; the raw value is kept in `orig_id`. Each assembly's available mass starts as the sum of its reported nuclide masses, and a running total tracks the material remaining in the facility.

Three composition fractions used by the selection policies are computed once at load time. For assembly a with total mass m_a and nuclide mass $m_{i,a}$ for nuclide i , the fissile mass fraction is

$$f_{\text{fissile},a} = \frac{m_{^{233}\text{U},a} + m_{^{235}\text{U},a} + m_{^{239}\text{Pu},a} + m_{^{241}\text{Pu},a}}{m_a}, \quad (3.1)$$

where the numerator sums the masses of ^{233}U , ^{235}U , ^{239}Pu , and ^{241}Pu in assembly a , and m_a is its total mass. The total plutonium mass fraction is

$$f_{\text{Pu},a} = \frac{\sum_{i \in \text{Pu}} m_{i,a}}{m_a}, \quad (3.2)$$

where the sum runs over all plutonium isotopes present in assembly a . The minor actinide mass fraction is

$$f_{\text{MA},a} = \frac{\sum_{i \in \text{Np}} m_{i,a} + \sum_{i \in \text{Am}} m_{i,a} + \sum_{i \in \text{Cm}} m_{i,a}}{m_a}, \quad (3.3)$$

where each sum runs over all isotopes of neptunium, americium, or curium in assembly a . These fractions are precomputed so the selection policies need not traverse the full nuclide composition on every request. The required inputs are the output commodity, `outcommod`, and the assembly data directory, `data_dir`. An optional maximum throughput, `throughput_kg`, in kilograms per time step, is effectively unlimited by default but can be reduced to represent a constraint on inventory retrieval, transportation, or processing.

3.3.2 Assembly Provenance and Geographic Coordinates

Preserving assembly provenance connects inventory selection results to later transportation, siting, and infrastructure analyses. The STANDARDS records identify each assembly's facility and storage location but not its coordinates, so an optional SQLite database associates a facility name with latitude and longitude. The database path is supplied through `coordinates_file`. If it is empty, the archetype still runs but every assembly keeps the default pair (0,0). When a database is provided, it is opened read-only and queried for the facility name, taking an exact match where one exists and otherwise a prefix match. All assemblies from a facility share one coordinate pair, cached after the first lookup. If no entry matches, the archetype warns and leaves that facility at (0,0).

Every completed transfer is recorded in a custom CYCLUS output table, `us_inventorySupply`, with one row per supplied trade (Table 3.2). The transferred material separately carries the composition built from the selected assembly, so a transaction can be traced to both its source assembly and its material composition.

3.3.3 Assembly Selection Policies

The archetype's principal methodological feature is its ability to choose among individual assemblies by explicit ranking policies. In resource exchange the solver decides which offers and requests match, but the supplier still chooses which physical resource fulfills an accepted trade; `us_inventory` uses this selection step to model different assumptions about access to the legacy inventory.

Table 3.2: Columns written to the `us_inventorySupply` output table.

Column	Description
Agent ID	Identifier of the supplying agent.
Time	Simulation time step of the transfer.
Facility	Source facility name.
Storage location	Source storage location name.
Storage type	Wet or dry storage.
Assembly ID	Original assembly identifier.
Initial uranium	Initial uranium mass of the assembly.
Latitude, longitude	Coordinates of the source facility.
Quantity	Mass supplied in the trade.
Selection policy	Policy in force for the simulation.

The policy is set through `selection_policy` and checked at initialization, where an unrecognized value stops the simulation with a descriptive input error. For each trade the archetype scans the eligible assemblies and keeps the one ranked highest under the active criterion, skipping empty assemblies and assemblies without a valid composition. Table 3.3 lists the 15 policies.

Table 3.3: Assembly selection policies implemented by the `us_inventory` archetype. Each policy ranks the eligible assemblies before a trade is fulfilled.

Policy	Assembly selected
<code>first</code>	First eligible assembly in the database.
<code>older</code>	Earliest discharge year.
<code>newer</code>	Latest discharge year.
<code>highest_burnup</code>	Largest reported maximum burnup.
<code>lowest_burnup</code>	Smallest reported maximum burnup.
<code>highest_enrichment</code>	Largest initial enrichment.
<code>lowest_enrichment</code>	Smallest initial enrichment.
<code>highest_initial_uranium</code>	Largest initial uranium mass.
<code>lowest_initial_uranium</code>	Smallest initial uranium mass.
<code>highest_fissile</code>	Largest fissile fraction (Eq. 3.1).
<code>lowest_fissile</code>	Smallest fissile fraction.
<code>highest_plutonium</code>	Largest plutonium fraction (Eq. 3.2).
<code>lowest_plutonium</code>	Smallest plutonium fraction.
<code>highest_minor_actinide</code>	Largest minor actinide fraction (Eq. 3.3).
<code>lowest_minor_actinide</code>	Smallest minor actinide fraction.

Ranking keeps the first eligible assembly as the initial candidate and replaces it only when a later assembly is strictly better under the criterion, so ties retain their load order priority.

Two optional inputs bias the pool toward a particular source. `preferred_facility` names a specific facility, and `storage_type` restricts the pool to `wet` or `dry` storage. These filters are preferential rather than exclusive: the matching assemblies are searched first, and if they cannot supply a trade the rest of the inventory is searched, with a warning the first time this fallback occurs. This keeps the archetype consistent with the DRE, which expects a bidding facility to remain able to supply an accepted trade.

Whether a request may be partially filled is governed by `allow_partial`. Selection first requires that an assembly cover the full requested mass; only if none can, and `allow_partial` is true, is a smaller quantity permitted. Each accepted trade is drawn from a single assembly. The archetype does not combine undersized assemblies, which preserves a one-to-one relationship between each trade response and the assembly

recorded in `us_inventorySupply`.

3.3.4 Material Bidding, Trading, and Accounting

Bidding and trading are handled in two stages: the facility first advertises how much material it can offer, and then, once trades are awarded, withdraws a specific assembly for each one.

When there are requests on the output commodity, the facility bids. Because the source filters are preferential, the whole remaining inventory is eligible, so the facility can fall back to nonmatching assemblies.

The compositions of the eligible assemblies are combined into a single mass-weighted bid composition,

$$\mathbf{x}_{\text{bid}} = \frac{\sum_{a \in \mathcal{E}} m_a \vec{\mathbf{m}}_a}{\sum_{a \in \mathcal{E}} m_a}, \quad (3.4)$$

where $\mathcal{E}$ is the set of eligible assemblies, m_a is the remaining mass of assembly a , and $\vec{\mathbf{m}}_a$ is its vector of nuclide mass fractions. The maximum mass offered is

$$M_{\text{offer}} = \min(M_{\text{eligible}}, M_{\text{throughput}}), \quad (3.5)$$

where $M_{\text{eligible}} = \sum_{a \in \mathcal{E}} m_a$ is the total mass of the eligible pool and $M_{\text{throughput}}$ is the configured throughput per time step. This value enters the portfolio as a capacity constraint. With partial fulfillment disabled, the archetype does not bid on a request larger than it can offer.

The bid composition represents the available pool but does not replace selection of the source assembly according to the active policy. After the solver awards trades, each one is matched to a source assembly by the procedure in Section 3.3.3, so the material returned is built from the selected assembly's composition, not the averaged bid composition.

Each accepted trade is limited by the requested quantity, the selected assembly's remaining mass, and the throughput remaining in the step, and fulfillment stops once the throughput or the inventory is exhausted. After each response the archetype updates

$$m_{a,t+1} = m_{a,t} - m_{\text{supplied}}, \quad (3.6)$$

$$M_{\text{inv},t+1} = M_{\text{inv},t} - m_{\text{supplied}}, \quad (3.7)$$

$$R_{t+1} = R_t - m_{\text{supplied}}, \quad (3.8)$$

where $m_{a,t}$ is the available mass of the selected assembly a , $M_{\text{inv},t}$ is the total mass remaining in the facility, R_t is the throughput remaining in the current step, and m_{supplied} is the mass delivered by the trade.

The remaining mass of each assembly is also persisted as archetype state, with one value stored for every loaded assembly. Restarting from a checkpoint reloads the assembly metadata and compositions from the source files, overwrites their initial masses with the persisted values, and rebuilds the inventory total from them. The simulation is stopped if the number of persisted values does not match the number of loaded assemblies, since that indicates the data directory changed after the checkpoint.

After each successful transaction the archetype appends a row to `us_inventorySupply`, giving a time-dependent history of which assemblies were selected, where they originated, how much they supplied,

and which policy produced that sequence. These records form the basis for the policy comparisons in Chapter 4.

3.4 The ads Archetype

The archetype represents an ADS operating as a transuranic burner. It receives an actinide-bearing feed from upstream facilities, accumulates one core loading, irradiates it for a prescribed cycle, and discharges the surviving actinides together with the fission products generated during the cycle. Like `us_inventory` (Section 3.3), it inherits from `cyclus::Facility`. Unlike `us_inventory`, it does not use the `CommodityProducer` toolkit class; instead, it explicitly requests feed, bids its discharged product, advances an irradiation cycle each time step, and participates in the DRE on both the request and bid sides. Its input feed and output commodity streams are illustrated in Figure 3.2.

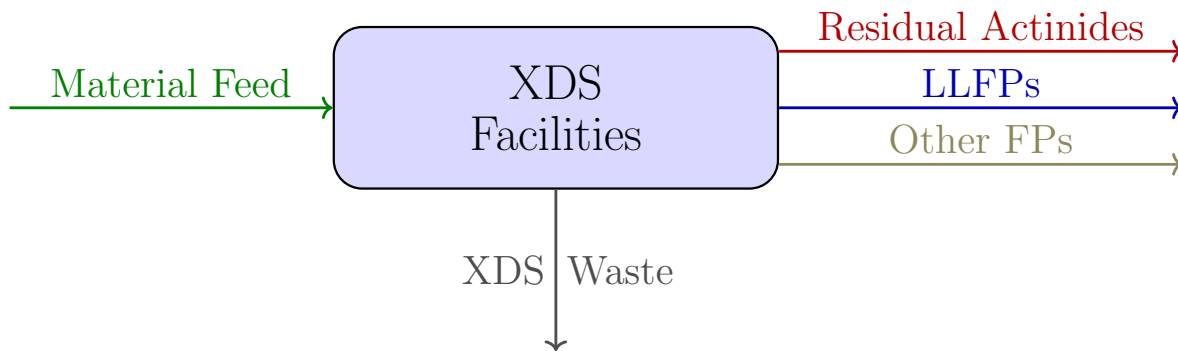


Figure 3.2: Structure of the XDS facility archetype within the CYCLUS framework, showing the input material feed and the output streams containing residual actinides, long-lived fission products, other fission products, and waste.

3.4.1 Reduced-Order Transmutation Model

Solving neutron transport and depletion throughout a fuel cycle simulation that spans several decades, for every cycle of every burner, is computationally intractable. The `ads` archetype therefore uses a *reduced-order* model: the physics is computed once, offline, by high-fidelity depletion calculations, and at runtime the archetype performs only a case lookup and an interpolation. This mirrors the recipe-based treatment of irradiation established in CYCLUS [10].

The offline physics is a library of SERPENT depletion cases (Section 3.4.2), each recording the time evolution of the full nuclide inventory of a fixed loading irradiated at constant power. The library is loaded once when the agent enters the simulation and held in memory.

Buffers and cycle. The facility holds three material buffers: `feed` accumulates received material, `core` holds the loading under irradiation, and `product` holds discharged material awaiting trade. The feed and core buffers share the `capacity` parameter, the mass of one core, while the product buffer is unbounded. Operation is a two-state batch cycle keyed on whether the core is occupied. An empty core is refueling; once `refuel_time` has elapsed and the feed holds a full core, one `capacity` of feed is loaded. An occupied core irradiates until its cycle counter reaches `cycle_time`, at which point it is discharged. The composition

is not altered during irradiation. Instead, the entire transformation is applied as a single transition at discharge because the core does not participate in resource exchange while loaded. Two persisted counters track progress through the irradiation and refueling periods so the operating state survives a simulation snapshot.

Feed requests. Feed may be requested on several input commodities listed in `incommods`, with an optional preference assigned to each commodity. The archetype requests the available space in its feed buffer, and the requests for the different commodities are mutually exclusive alternatives in one portfolio. It therefore draws one core load from whichever supplier the exchange prefers rather than the full amount from each. An optional `feed_recipe` may be attached; when it is empty, the request is for blank material and the archetype accepts whatever composition the supplier offers. This is the mode used here, since the feed composition varies with the inventory selection policy and with any upstream separations.

Simulation time to irradiation time. CYCLUS advances in time steps, whereas the SERPENT cases are tabulated in irradiation days. The parameter `days_per_step`, Δ_d , converts between them, so the irradiation time read at discharge is

$$t_{\text{irr}} = n_{\text{cycle}} \Delta_d, \quad (3.9)$$

where n_{cycle} is the cycle length in time steps (`cycle_time`). If Δ_d is not set, it is derived from the time step duration. This is what gives the cycle length physical meaning: a longer cycle addresses a later, more-depleted point on the trajectory.

Case selection. At discharge the core is matched to the library case that best represents it. The core is reduced to the mass fractions of three actinide groups, normalized to the total actinide mass:

$$f_{\text{MA}} = \frac{m_{\text{NP}} + m_{\text{Am}} + m_{\text{Cm}}}{m_{\text{act}}}, \quad f_{\text{Pu}} = \frac{m_{\text{Pu}}}{m_{\text{act}}}, \quad f_{\text{U}} = \frac{m_{\text{U}}}{m_{\text{act}}}, \quad (3.10)$$

where each element mass (m_{NP} , m_{Am} , m_{Cm} , m_{Pu} , m_{U}) is summed over all isotopes of that element, $m_{\text{act}} = \sum_{Z \geq 89} m$ is the total actinide mass, and the minor actinides are neptunium, americium, and curium. Nuclides below $Z = 89$ are excluded, so fission products in recycled feed do not affect selection. The same fractions are computed for each case from its initial ($t = 0$) actinide vector. Each case c is scored by

$$S_c = w_P \frac{|P - P_c|}{\max(P, P_c)} + \sum_{g \in \{\text{MA}, \text{Pu}, \text{U}\}} |f_g - f_{g,c}|, \quad (3.11)$$

where P and P_c are the facility and case powers, f_g and $f_{g,c}$ are the core and case group fractions, and $w_P = 10$ weights the power term. The case with the lowest score is selected; the large w_P makes the power tier dominate, so a core is not matched across power levels on composition alone. A single nearest case is chosen. There is no averaging across cases, and interpolation occurs only along that case's time axis.

Composition transformation. The selected case gives nuclide masses $m_n(t_k)$ at tabulated times t_k . For the interval $[t_{\text{lo}}, t_{\text{hi}}]$ bracketing t_{irr} , each nuclide is linearly interpolated,

$$m_n(t_{\text{irr}}) = (1 - \alpha) m_n(t_{\text{lo}}) + \alpha m_n(t_{\text{hi}}), \quad \alpha = \frac{t_{\text{irr}} - t_{\text{lo}}}{t_{\text{hi}} - t_{\text{lo}}}, \quad (3.12)$$

over every actinide and fission product the case tracks, and the result becomes the discharged composition. Outside the tabulated range the composition is clamped to the nearest endpoint rather than extrapolated: depletion saturates as the most fissile nuclides are consumed, so a straight-line projection past the last point would overpredict burnup and eventually yield negative masses. A cycle that routinely runs past the range signals a misconfigured time mapping or a need for longer depletion runs, not a case for extrapolation.

Mass balance. The transformed composition holds both surviving actinides and the fission products made during irradiation, so mass is conserved to the extent SERPENT tracks it. With $M(t)$ the total tracked (actinide plus fission product) mass of the case at time t , the discharged mass scales from the loading M_{in} by

$$R = \min\left(1, \frac{M(t_{\text{irr}})}{M(0)}\right), \quad M_{\text{out}} = R M_{\text{in}}, \quad (3.13)$$

the upper limit of unity prevents a ratio slightly greater than one from creating mass. The residual $(1 - R) M_{\text{in}}$ leaves the simulation as a recorded mass defect, representing mass not retained in the SERPENT output tables, including mass converted to energy during fission and any light or otherwise untracked nuclides. Writing $a_c(t) = M_{\text{act},c}(t)/M_c(t)$ for the actinide fraction of the matched case, the destroyed actinide mass reported for the cycle is

$$\Delta M_{\text{act}} = M_{\text{in}} a_c(0) - M_{\text{out}} a_c(t_{\text{irr}}). \quad (3.14)$$

Equation 3.14 measures the net reduction in total actinide mass represented by the matched depletion case. Both $a_c(0)$ and $a_c(t_{\text{irr}})$ include the complete actinide inventory reported in the Serpent actinide mass table. Consequently, conversion from one actinide to another, for example, conversion of a minor actinide into plutonium, does not count as destruction, because the product nuclide remains in the discharged actinide inventory. This is a summary quantity only; the discharged isotopics come directly from Equation 3.12. Discharged material is then offered on the output commodity under a capacity constraint equal to the product mass, and accepted trades draw from the product buffer for routing to storage, separations, or disposal.

3.4.2 Serpent Depletion Cases

These depletion cases were produced at UIUC as part of the EINSTEIN project by Runxia Wen, of the neutronics analysis group led by Professor Tomasz Kozłowski. This section documents the model they represent so that the archetype’s inputs can be interpreted. Each case is a continuous-energy Serpent 2 Monte Carlo depletion calculation [48] that tracks the full nuclide inventory (`set inventory all`), producing the time-dependent actinide and fission product mass tables the archetype later reads; the material volumes required for depletion normalization are obtained by a Monte Carlo volume calculation.

Figure 3.3 shows the model geometry, an axisymmetric cylindrical ADS core built from concentric z -cylinders. Its principal regions and dimensions are summarized in Table 3.4. A central spallation target filled with liquid lead sits on the core midplane, surrounded by a thin structural target wall and, above it, an evacuated beam duct through which the proton beam reaches the target. The fuel occupies the annular region around the target tube, and the whole assembly is enclosed in structural material and an outer container. A vacuum (black) boundary condition is applied at the outermost surface, so neutrons leaking

past the container are lost. The single target, single homogeneous fuel region, and absence of detail at the assembly or pin level are consistent with the reduced-order role of these cases: they are intended to characterize the bulk transmutation behavior of one core load of feed rather than its detailed spatial power distribution.

Table 3.4: Principal regions of the Serpent ADS model. Dimensions are taken directly from the input decks; all cylinders are coaxial about the z -axis.

Region	Radius (cm)	Axial span (cm)	Material
Spallation target	0–17.5	150–210	Liquid lead
Beam duct (vacuum)	0–17.5	210–400	Void
Target wall	17.5–18.5	149–400	Structural
Fuel	18.5–150	0–300	Minor actinide feed
Container / structure	150–250	–100–400	Structural

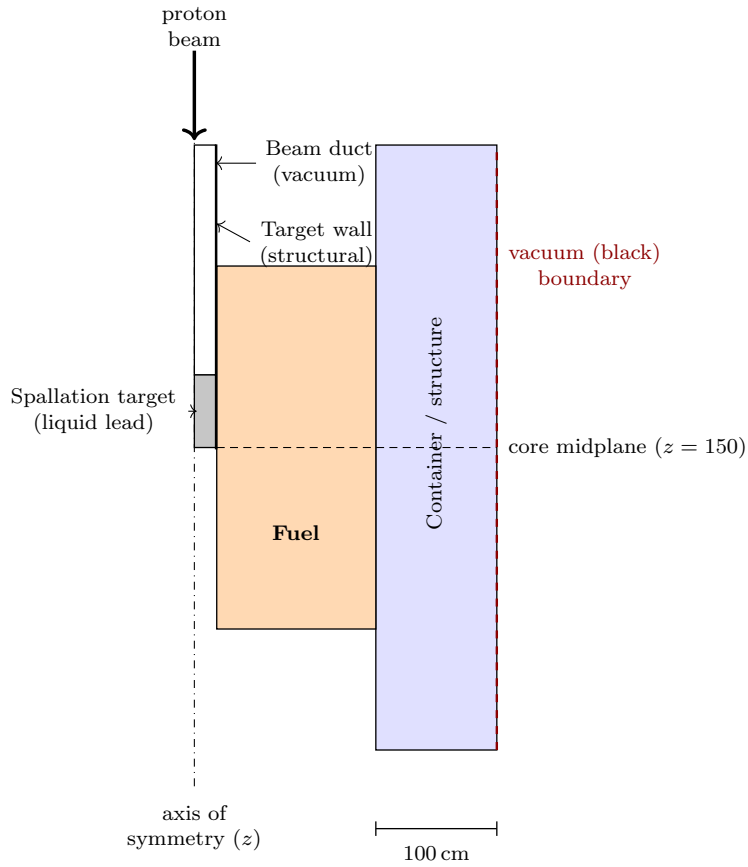


Figure 3.3: To-scale half cross-section (r – z) of the axisymmetric Serpent ADS model, mirrored about the axis of symmetry. Concentric regions and dimensions are taken from Table 3.4: a central spallation target filled with liquid lead on the core midplane, an evacuated beam duct above it, a thin structural target wall, the surrounding fuel annulus, and the outer container. A vacuum (black) boundary condition is applied at the outermost surface.

Fuel Compositions and the Case Set

The cases differ in the composition loaded into the fuel region, which is what makes them a *library* rather than a single run. Each fuel is a transuranic feed dispersed in a lead matrix. Two representative cases

illustrate the range spanned by the library. In one, the feed is essentially pure minor actinide dispersed in lead; in another, the feed contains equal masses of plutonium and minor actinides in lead. These two feeds bracket the transmutation regimes of interest: a charge containing only minor actinides maximizes the minor actinide mass destroyed in each cycle but presents the greatest reactivity challenge during loading, whereas adding plutonium reduces that challenge while shifting some of the destruction burden onto bred and fissioned plutonium. The initial actinide mass of each case, taken from the day-zero row of its actinide mass table, is the quantity the archetype’s **capacity** parameter should reproduce, so that a loaded core matches the case it is compared against.

Each case is depleted along a common “Day [d]” time axis. The irradiation history is built up from a sequence of chained restart runs: an initial segment resolves the first fraction of a day with fine steps (for example, cumulative times of 0.001, 0.1, 0.2, and 0.5 d) to capture the rapid early transient in short-lived fission products, and each subsequent segment reads the binary restart file written by the previous one and advances the total burnup further (for example, from 730 to 760 d). Reassembled, these segments give the continuous nuclide inventory versus irradiation time that the archetype consumes: an actinide mass table (`plot_data_15_actinide_masses_full.txt` or `plot_data_06_actinide_masses.txt`), a fission product mass table (`plot_data_14_fission_product_masses.txt`) on the same axis, and a power history (`plot_data_03_power_vs_time.txt`) used for case matching. The archetype reads a case’s composition at the irradiation time implied by the cycle length by linearly interpolating these tables along the day axis, so the resolution of the discharged composition is set by the density of the Serpent time grid rather than by the CYCLUS time step.

3.4.3 Irradiation Event Recording

Besides the standard CYCLUS tables, the archetype writes a custom **AdsEvents** table so each cycle can be reconstructed without parsing the full isotopic history. Every row carries the agent ID, time, event type, matched case, and a mass value (Table 3.5); the matched case is blank on a load, since matching happens at discharge.

Table 3.5: Event types recorded in the **AdsEvents** output table.

Event	Recorded value
Load	Mass loaded from feed into the core.
Discharge	Core mass entering the transformation, M_{in} , before Equation 3.13; the matched case is recorded here.
ActinideDestroyed	Destroyed actinide mass, Equation 3.14.
FissionProducts	Fission product mass in the discharged material.
MassDefect	Mass not retained, $(1 - R) M_{\text{in}}$.

The three discharge rows are written only when positive. The **ActinideDestroyed** series is the primary performance metric, and its cumulative sum gives the actinide mass a facility removes from the fuel cycle. With the standard CYCLUS **Transactions** and **ExplicitInventory** tables, these events close the actinide balance: the actinide initially present equals the sum destroyed, held in buffers, and resident downstream.

3.5 Simulation Scenarios

The two archetypes developed in this chapter are exercised in a set of fuel cycle simulations that share one flowsheet and differ along three axes: the composition burned in the ADS, the length of the simulated horizon, and whether the discharged material is recycled. Two burner configurations are used, corresponding to the two representative Serpent cases described in Section 3.4.2: a core containing only minor actinides (ADS 1) and a core containing equal masses of plutonium and minor actinides (ADS 2). Each configuration is simulated over 30-year and 100-year horizons in both once-through and recycling modes, resulting in eight simulations.

3.5.1 Fuel Cycle Flow

Every simulation uses the same chain of facilities. The `us_inventory` facility supplies discharged assemblies on the `snf` commodity under the `highest_minor_actinide` selection policy (Section 3.3.3), so that the assemblies richest in neptunium, americium, and curium are drawn first. A `CYCAMORE Separations` facility partitions this material: the minor actinides are routed to the burner feed, while uranium, plutonium (except in ADS 2), and the fission products fall to a leftover stream sent to a waste `Sink`. The ADS irradiates the separated feed and discharges the surviving actinides together with the fission products produced during the cycle. Depending on the scenario, that discharge is either sent to a `Sink` (once-through) or returned to `Separations` (recycling).

The two burner configurations shape the feed differently. For ADS 1, which contains only minor actinides, `Separations` extracts a single stream of neptunium, americium, and curium at 99% efficiency and feeds it directly to the ADS. For ADS 2, which contains equal masses of plutonium and minor actinides, the feed cannot be taken straight from separations: the U.S. inventory holds roughly ten times more plutonium than minor actinide, so a stream drawn from it is far from an equal split. `Separations` therefore produces two streams, one of minor actinides and one of plutonium, and a `CYCAMORE Mixer` blends them in a fixed 0.5/0.5 mass ratio to reproduce the ADS 2 Serpent feed. The minor actinide stream limits the feed rate, so surplus plutonium accumulates in its buffer. This limitation arises from the composition of the inventory rather than from an imposed modeling constraint.

The separations facilities in these scenarios are idealized material partitioning models. The simulations assume that an operating reprocessing system can recover plutonium and the minor actinides into distinct product streams at the specified 99% efficiencies. This is an enabling boundary condition of the analysis rather than a prediction of deployable process performance. The `CYCAMORE Separations` archetype applies user defined partitioning efficiencies but does not represent process chemistry, product purity, decontamination factors, plant availability, secondary waste generation, remote fabrication of actinide bearing fuel, safeguards, licensing, or the broader policy constraints associated with commercial reprocessing. The results therefore describe the material flow performance of the ADS scenarios conditional on such separations infrastructure being available.

The recycling and once-through modes differ only in the routing of the burner discharge. In once-through mode the discharge is a terminal waste stream held in a `Sink`. In recycling mode it is returned to `Separations`, which recovers the surviving minor actinides, and, for ADS 2, plutonium, into fresh feed, so that actinides make repeated passes through the burner while uranium and fission products are removed to waste on each pass.

The two routings are shown in Figure 3.4 for the once-through cycle and Figure 3.5 for the recycling cycle.

All simulations share the remaining settings. Time advances in monthly steps of 30.4375 d from a start of January 2025, radioactive decay is evaluated lazily over the horizon, and explicit inventory tracking is enabled. The inventory, separations, and mixing facilities each operate at a throughput of 250,000 kg per step, large enough that material handling does not throttle the burner.

3.5.2 Simulation Runs

The two burner configurations fix the ADS and separations parameters summarized in Table 3.6. Each core mass matches the day-zero actinide mass of the corresponding Serpent case, so a loaded core maps onto the case it is compared against; the cycle lengths differ by more than the mass alone, as the large core containing only minor actinides is irradiated for ten years per cycle while the smaller 50/50 core is cycled every two years.

Table 3.6: ADS and separations configuration for the two burner cases. Cycle and refueling times are given in monthly time steps, with the equivalent duration in parentheses.

Parameter	Case 3 (MA only)	Case 4 (50/50 Pu-MA)
Initial core mass	27,000 kg	4,418 kg
Cycle length	120 steps (10 yr)	24 steps (2 yr)
Refueling time	6 steps (6 mo)	4 steps (4 mo)
Power level	3000 MW	3000 MW
Separated streams	Np, Am, Cm	Np, Am, Cm and Pu
Feed shaping	Fed directly	CYCAMORE Mixer, 0.5/0.5
Serpent case	892_vt_ma_Pb	1.3_vt_5050_puma_Pb

Each configuration is simulated over both horizons and in both modes, giving the eight runs listed in Table 3.7. Across most of these runs the once-through and recycling modes produce nearly identical histories, because the burner is limited by its cycle duration rather than by feed availability. Within the horizon, it completes the same number of cores whether or not the discharge is recycled, and the recovered material only displaces inventory that was still available. The exception is ADS 1 over the 100-year horizon. Its large core load and ten-year irradiation cycle draw down the accessible minor actinide inventory, and recycling allows operation to continue beyond what the once-through supply can support. The comparison between recycling and once-through in Chapter 4 therefore focuses on that pair of runs; for the remaining cases a single representative run is shown, since the two modes coincide.

Table 3.7: Simulation runs. Each burner configuration is run over both horizons and in both fuel management modes.

Run	Fuel case	Mode	Horizon	Steps
1	MA only	Once-through	30 yr	360
2	MA only	Recycle	30 yr	360
3	MA only	Once-through	100 yr	1200
4	MA only	Recycle	100 yr	1200
5	50/50 Pu-MA	Once-through	30 yr	360
6	50/50 Pu-MA	Recycle	30 yr	360
7	50/50 Pu-MA	Once-through	100 yr	1200
8	50/50 Pu-MA	Recycle	100 yr	1200

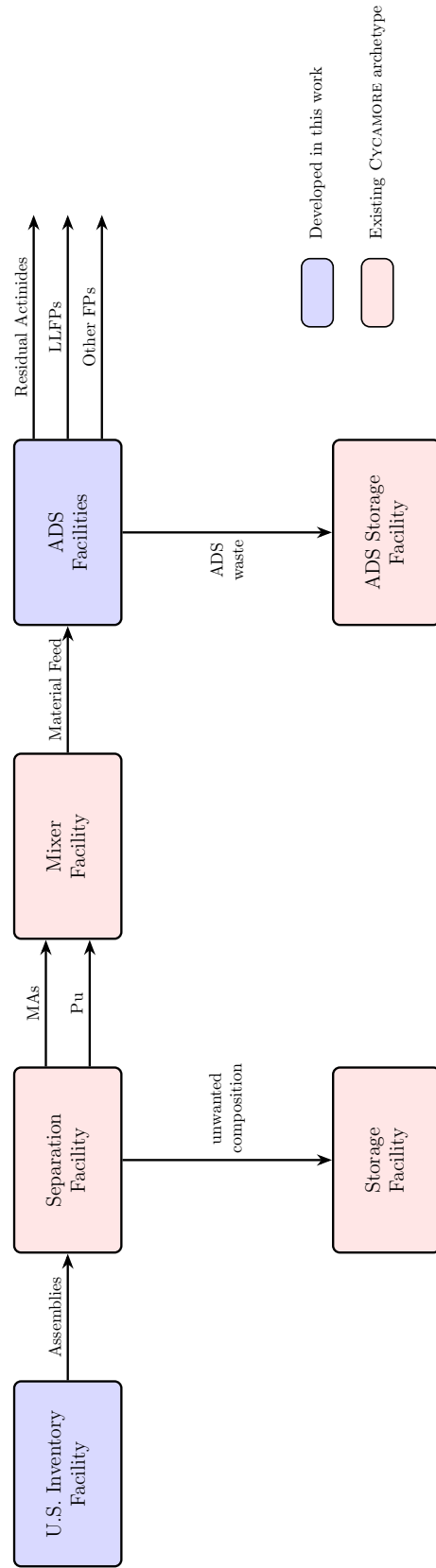


Figure 3.4: Conceptual flow for the once-through fuel cycle. Material from the STANDARDS 5.0.1 inventory is separated, blended when needed, and irradiated in the ADS. Separation waste and ADS discharge products follow separate terminal storage paths. Blue facilities denote archetypes developed for this work, while pink facilities represent existing CYCAMORE archetypes.

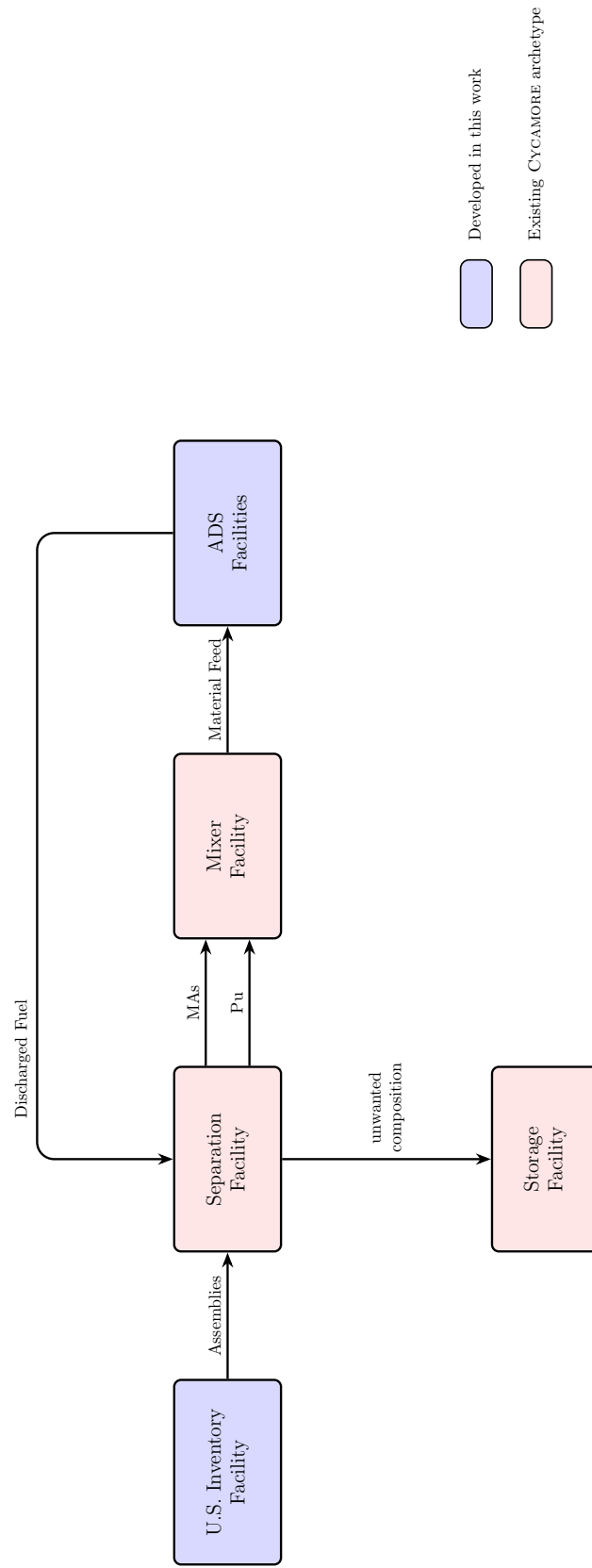


Figure 3.5: Conceptual flow for the closed (recycling) fuel cycle. Material from the STANDARDS 5.0.1 inventory is separated, blended when needed, and irradiated in the ADS; the discharged fuel is then returned to the separation facility for recovery and reuse. The unwanted separation stream is sent to storage. Blue facilities denote archetypes developed for this work, while pink facilities represent existing CYCAMORE archetypes.

Chapter 4

Results

This chapter presents the results produced by the `us_inventory` and `ads` archetypes described in Chapter 3, following the material from the inventory that supplies the fuel cycle through the transmutation it enables to the waste it leaves behind. It begins by characterizing the U.S. spent fuel inventory as loaded by `us_inventory`, confirming that the STANDARDS 5.0.1 dataset is ingested correctly and establishing the scale of material available. It then examines how the choice of selection policy shapes the feed supplied to the burner, both in its delivery over time and in its geographic provenance, before turning to the transmutation performance of the ADS: the actinide destroyed per cycle, the two burner configurations considered, and the effect of recycling. Attention then moves to the fate of the material once it has been processed, describing the composition of the remaining inventory and the resulting waste stream. The chapter closes by drawing these results together and interpreting what they imply for a minor actinide transmutation mission.

4.1 Inventory Characterization

Before any transmutation results are presented, this section characterizes the inventory that the `us_inventory` archetype ingests, establishing both that the STANDARDS 5.0.1 dataset is loaded correctly and the scale and distribution of the material available to the fuel cycle. Because the inventory load is deterministic, the same directory of assembly records always produces the same pool. Therefore, these results depend only on the input data and the parser, not on the simulation dynamics.

Reading the 75 assembly record files yields 315,976 individual assemblies, totalling 8.69×10^4 t of heavy metal, drawn from 61 facilities for which geographic coordinates are available. Summing the minor actinide content of every assembly, as defined in Equation 3.3, gives a total accessible minor actinide inventory of 103.8 t. Minor actinides account for only 0.12% of the total heavy metal because discharged light water reactor fuel contains relatively little neptunium, americium, and curium. In terms of burner loading, the entire national minor actinide inventory amounts to fewer than four 27 t ADS 1 cores (Table 3.6). This scarcity is the central constraint on the transmutation mission and recurs throughout the results: the burner is limited far more by the availability of minor actinides than by any reactor parameter.

Figure 4.1 shows the geographic distribution of this material, with each facility placed at its coordinates and shaded by the mass of spent fuel it supplies over the full simulation horizon. Because the run draws down the accessible inventory, the drawn mass approaches the available inventory at each site, so the fig-

ure doubles as a map of where the national inventory resides. The distribution is strongly weighted toward the eastern and south-central United States, where the largest single sites each supply on the order of 2,500–2,700 t, while the smaller western sites contribute a few hundred tonnes. This spatial concentration is the starting point for the provenance analysis of Section 4.2, since any policy that draws assemblies in a particular order also implies a particular geographic sequence of retrieval.

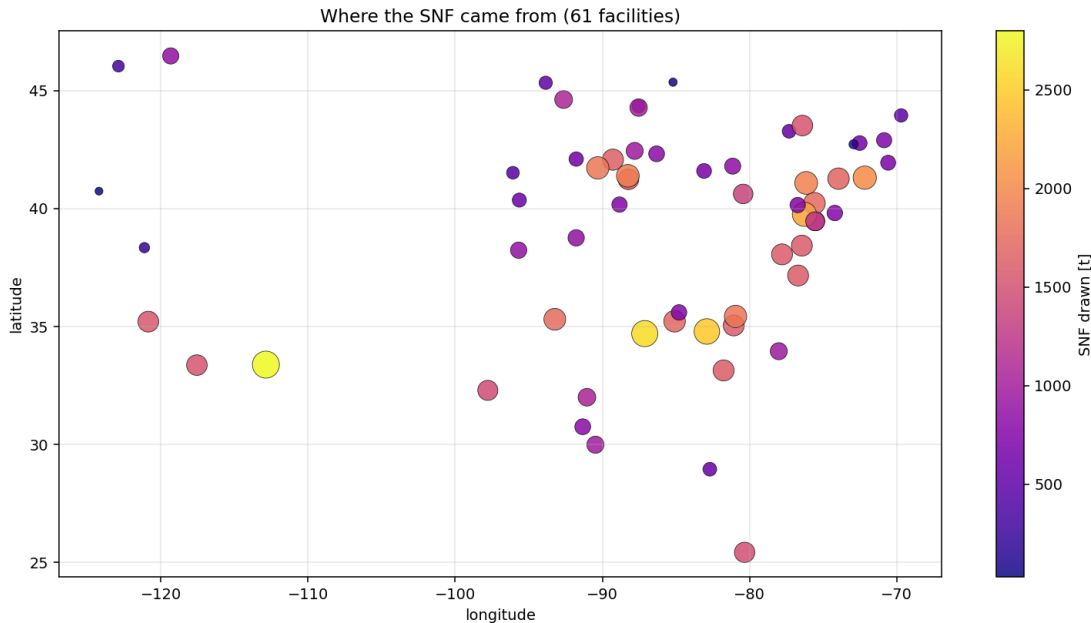


Figure 4.1: Geographic distribution of the spent nuclear fuel supplied by the `us_inventory` archetype. Each marker is one of the 61 facilities with known coordinates, positioned by latitude and longitude and shaded and sized by the mass of spent fuel drawn over the full simulation horizon. The inventory is concentrated in the eastern and south-central United States.

Establishing this baseline confirms that the archetype ingests the assembly-level dataset faithfully and quantifies the material the transmutation fuel cycle has to work with. The remainder of this chapter examines how that material is selected (Section 4.2), how it is transmuted in the ADS (Section 4.3), and how the accessible inventory evolves under recycling (Section 4.4).

4.2 Effect of Selection Policy on the Feed

The `us_inventory` archetype selects individual assemblies by an explicit ranking policy (Section 3.3.3), and that choice determines the mass and composition of material delivered to separations and, ultimately, to the burner. The selection mechanism acts on the static inventory pool and is deterministic in the assembly ordering it produces: it does not depend on downstream reactor dynamics. Its effect on the feed can therefore be characterized directly, from the inventory data and from standalone `us_inventory` runs, independently of the full fuel cycle, and that is how it is presented in this section. The production scenarios of Chapter 4 all use `highest_minor_actinide` (Section 3.5.1); the comparisons here show the benefit of that choice relative to the alternatives.

4.2.1 Delivery Trajectory

Because every policy eventually draws from the same finite pool, the total mass of minor actinides that any policy can deliver is identical: the cumulative curves all terminate at the same 103.8 t (Section 4.1). What the policy changes is the *trajectory*, or how quickly minor actinides accumulate as assemblies are drawn. Figure 4.2 shows the cumulative minor actinide delivered against the number of assemblies drawn, for the three representative policies of set A, with a horizontal line marking one 27 t ADS 1 core.

The separation is large. Ranking by `highest_minor_actinide` reaches a full core after roughly 4.8×10^4 assembly draws, whereas `lowest_minor_actinide` requires about 1.28×10^5 , a factor of 2.7 more material handled for the same core. The unranked `first` baseline falls between them, at about 7.7×10^4 draws, and its curve is visibly segmented: because `first` draws facility by facility in load order, its slope steps up and down as it moves through sites with different minor actinide contents, whereas the ranked policies produce smooth, monotonically curving trajectories. The timing here is expressed in assemblies drawn rather than calendar time, since the mapping to time depends on the retrieval throughput per time step, which is not the subject of this comparison.

This is the practical face of the scarcity established in Section 4.1. With the national inventory holding fewer than four cores of minor actinide in total, the order of retrieval governs how much spent fuel must be handled, and, as shown next, from how many sites, to assemble even a single core.



Figure 4.2: Cumulative minor actinide delivered as a function of assemblies drawn for three selection policies. All policies terminate at the same total because they draw from one finite pool; they differ only in how quickly minor actinides accumulate. The dashed line marks one 27 t ADS 1 core; markers indicate where each policy first supplies a complete core. `highest_minor_actinide` reaches a core with roughly 2.7 times fewer draws than `lowest_minor_actinide`.

4.2.2 Geographic Provenance of the Feed

Because the selection order is also a geographic order, each policy implies a different set of sites supplying the first core. Figure 4.3 shows the minor actinide contributed by each facility to the first 27 t core under the set A policies. The contrast is stark. The `first` policy, drawing in load order, concentrates the core in a handful of sites, each contributing on the order of 2 t; `highest_minor_actinide` instead spreads the draw across many facilities, drawing from assemblies with high minor actinide fractions wherever they reside, so that no single site dominates. Ranking for feed quality thus trades a geographically concentrated retrieval pattern for a distributed one. This tradeoff connects directly to transportation and logistics, which fall outside the scope of this work but are enabled by the provenance recorded in `us_inventorySupply` (Table 3.2).

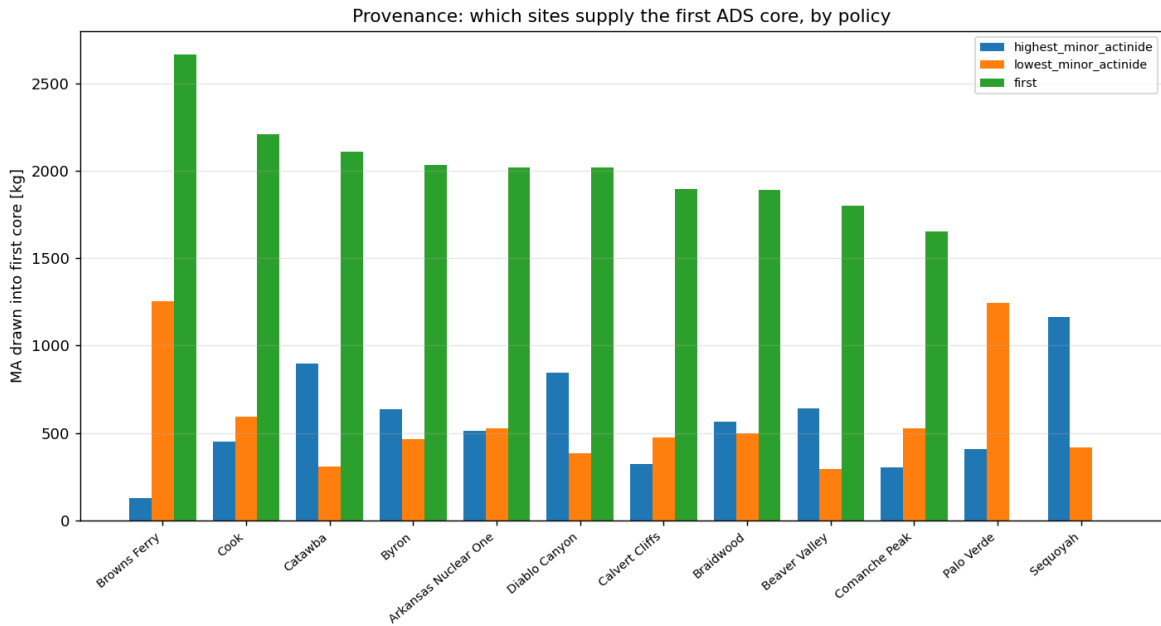


Figure 4.3: Minor actinide contributed by each facility to the first 27 t core, for the set A policies. `first` concentrates the core in a few sites drawn in load order, while `highest_minor_actinide` distributes it across many facilities.

The effect is not specific to ranking by minor actinide content. When the pool is instead ordered by discharge age (`older`, `newer`) or by reported burnup (`highest_burnup`, `lowest_burnup`), the set of sites supplying the first core shifts again, and each policy favors the facilities whose assemblies rank highest under its criterion. This confirms that the archetype’s selection acts as a general lever on feed provenance: any physical property recorded in the assembly data can be used to reshape both the composition and the geographic origin of the material entering the fuel cycle.

A standalone `us_inventory` run confirmed that the mechanism operates identically within a full CYCLUS simulation and in an offline analysis of the pool. The average composition of the supplied material followed the active policy. Policies that ranked assemblies by plutonium content or burnup supplied material with the correspondingly higher plutonium and fissile fractions.

4.3 ADS Transmutation Performance

This section examines how the ADS transmutes the separated feed. Of the eight runs defined in Section 3.5.2, most produce nearly identical histories. As anticipated there, once-through and recycling operation coincide whenever the burner is limited by its cycle duration rather than by feed availability, so one representative run is sufficient for those cases. The results therefore reduce to two comparisons. The first contrasts the core containing only minor actinides (ADS 1) with the core containing equal masses of plutonium and minor actinides (ADS 2) over the 100-year horizon, establishing how core size and feed composition govern burner operation. The second compares recycling with once-through operation for ADS 1, the only pairing in which the two modes diverge. For a given burner configuration, the transmutation result for each completed cycle is identical between modes and horizons; what changes is the number of cycles completed before the burner exhausts its feed.

4.3.1 Transmutation and Burn Fraction for Each Cycle

At each discharge the core composition is read from the matched SERPENT case (Section 3.4.1), giving the surviving actinides and the fission products produced during the cycle. Figures 4.4 and 4.5 show the cumulative fission product yield of the two cores as a function of fission product mass number. Both display the familiar double-humped distribution of low-to-moderate-energy actinide fission, with a light peak near $A \approx 100$ and a heavy peak near $A \approx 138$, separated by the symmetric fission valley. For ADS 1 (Figure 4.4), this signature appears even though the feed contains no uranium or plutonium, confirming that the minor actinides themselves undergo fission in the fast, subcritical core. The double-humped distribution arises from the shell structure of the fission fragments rather than from one particular actinide species. In both cases the three burnup snapshots nearly overlie one another, since the cumulative yield shape is set by the actinide mixture, which shifts only slowly with irradiation.

The two cores differ in the details of the distribution, and the difference reflects their feed compositions. ADS 2 contains a substantial plutonium fraction. Fission of ^{239}Pu proceeds through a heavier compound nucleus than fission of the actinides that dominate the ADS 1 feed. This shifts the light peak slightly upward in mass and partially fills the symmetric fission valley, reducing the peak-to-valley ratio. ADS 1, which contains a lighter mixture dominated by americium and neptunium, shows a marginally deeper valley and a light peak shifted slightly lower. The difference is subtle because both feeds fission in the same fast spectrum and span overlapping actinide masses, but it is the expected distinction between feeds with and without plutonium. It also confirms that the reduced-order model carries the difference in composition through to the fission product inventory.

The mass accounting for each cycle is summarized by the burn fraction, defined as the ratio of actinide mass destroyed to actinide mass charged. For ADS 1, this fraction remains constant at 41.3% across all cycles (Figure 4.6). This behavior follows directly from the reduced-order model: each core is irradiated to the same point on the same depletion trajectory, so every cycle removes the same fraction of its charge. For ADS 2, the burn fraction rises to just under 50% (Figure 4.7) because the plutonium in the feed fissions more readily than the actinides that dominate the ADS 1 feed.

The burn fraction reported here is the net reduction in total actinide mass, calculated from the `ActinideDestroyed` event according to Equation 3.14. Equation 3.13 first determines the mass retained in the discharged product, after which the archetype subtracts the total actinide mass remaining in that product from the total actinide mass in the matched case's initial loading. Because plutonium is included in the total actinide in-

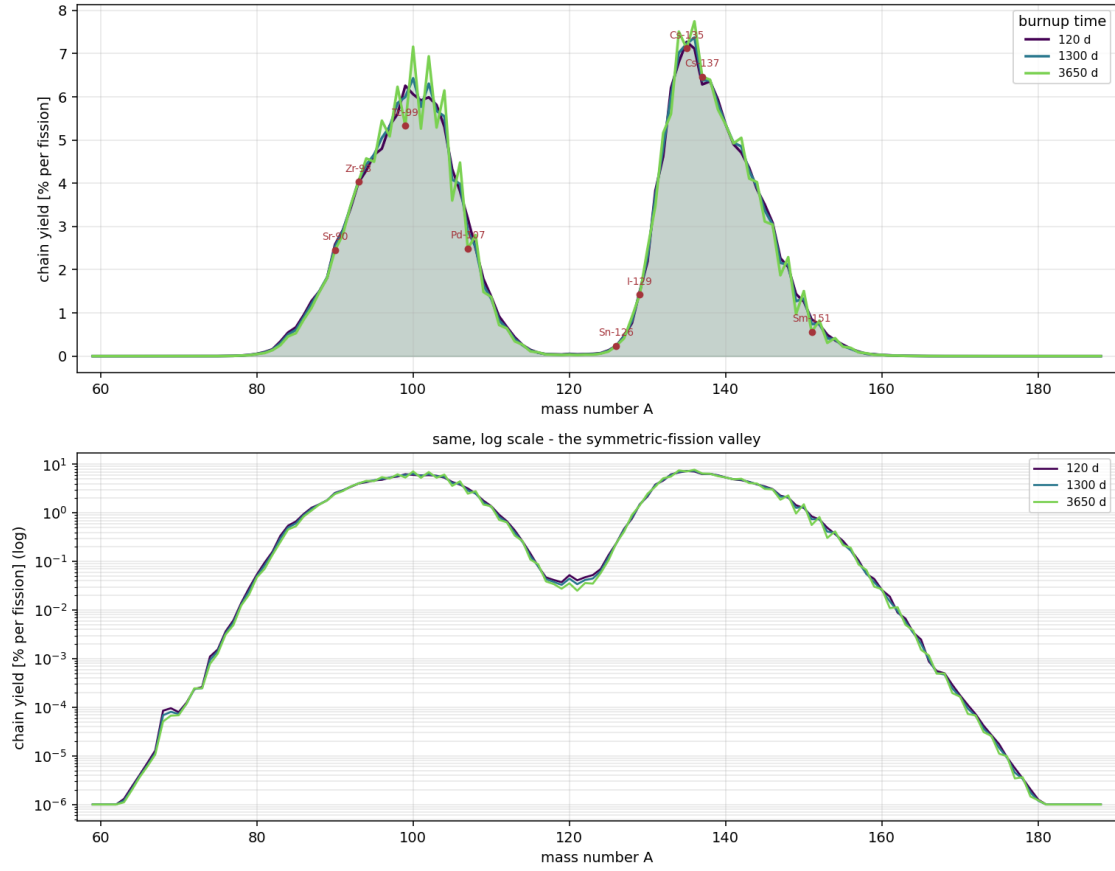


Figure 4.4: Cumulative fission product chain yield of ADS 1 as a function of fission product mass number at three points in the irradiation. The double-humped distribution appears despite the absence of uranium or plutonium in the feed, confirming that minor actinides are the fissioning species.

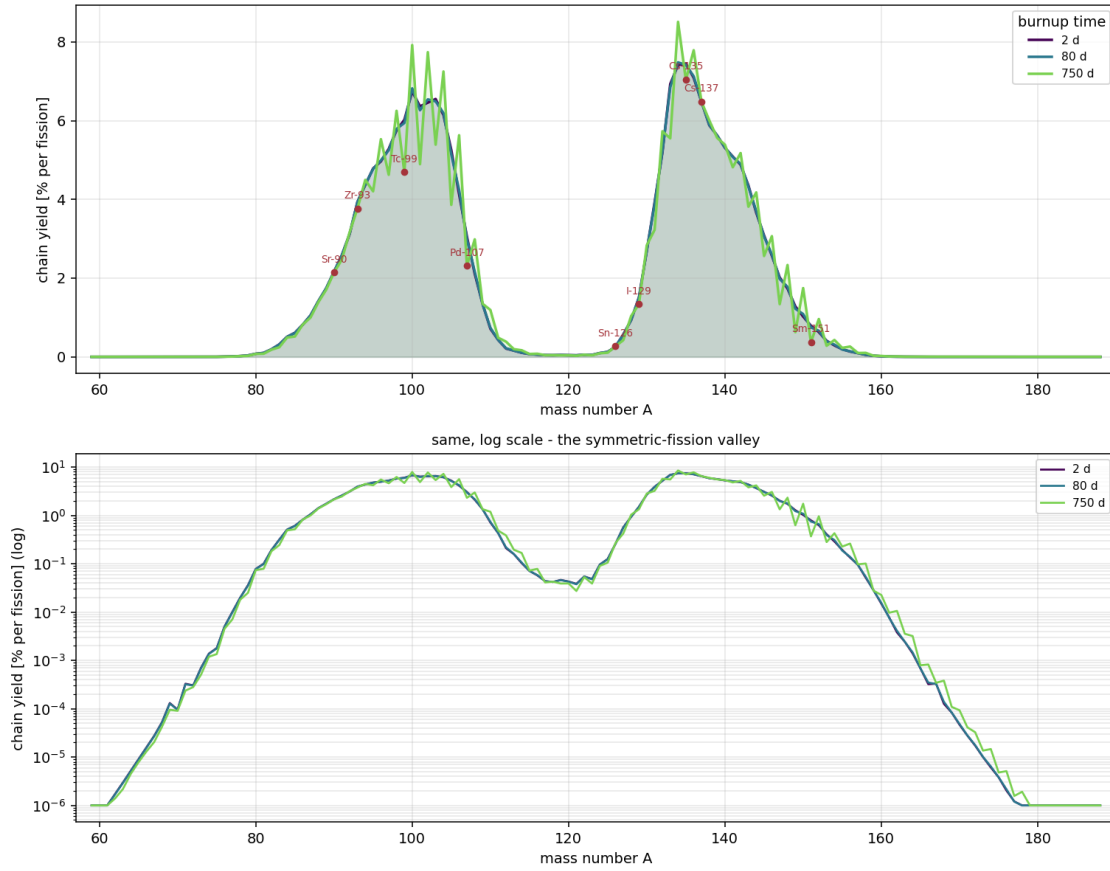


Figure 4.5: Cumulative fission product chain yield of ADS 2. Relative to ADS 1, the feed containing plutonium partially fills the symmetric fission valley and shifts the light peak slightly higher in mass, as expected for a heavier fissioning system.

ventory, minor actinides converted into plutonium remain in $M_{\text{out}}a_c(t_{\text{irr}})$ and are not counted as destroyed. The reported value therefore represents net actinide destruction in the tracked depletion inventory, rather than the disappearance of minor actinides alone.

This destruction rate can be checked against first principles. Fissioning one kilogram of actinide releases approximately 0.93 GW-d of thermal energy, so a 3000 MW_{th} core destroys about 3.2 kg of actinide per full-power day, or roughly 1,120 kg yr⁻¹ at full power. Applied to the ten-year ADS 1 cycle, this predicts destruction close to the 11,150 kg read from the SERPENT case, agreeing to within a few percent. The reduced-order transmutation model is therefore consistent with the fission energy balance, an independent validation of the archetype's mass accounting.

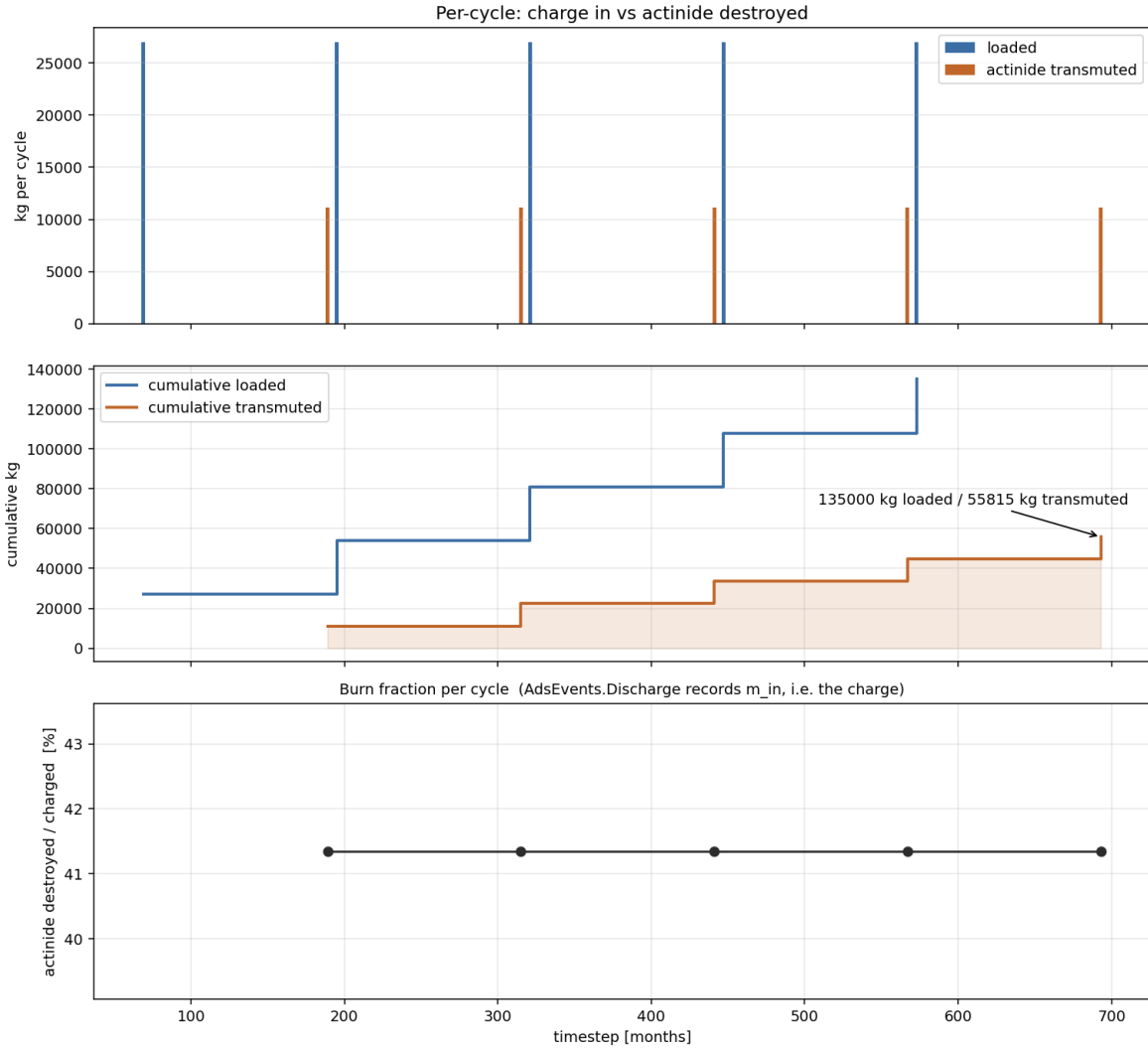


Figure 4.6: Charge and actinide destruction for each cycle of the 100-year ADS 1 recycling run. Each cycle charges 27 t and records 11.15 t of net actinide destruction, corresponding to a constant burn fraction of 41.3% in the bottom panel. The cumulative panel at the center reaches 135 t charged and 55.8 t of cumulative net actinide destruction over five cycles.

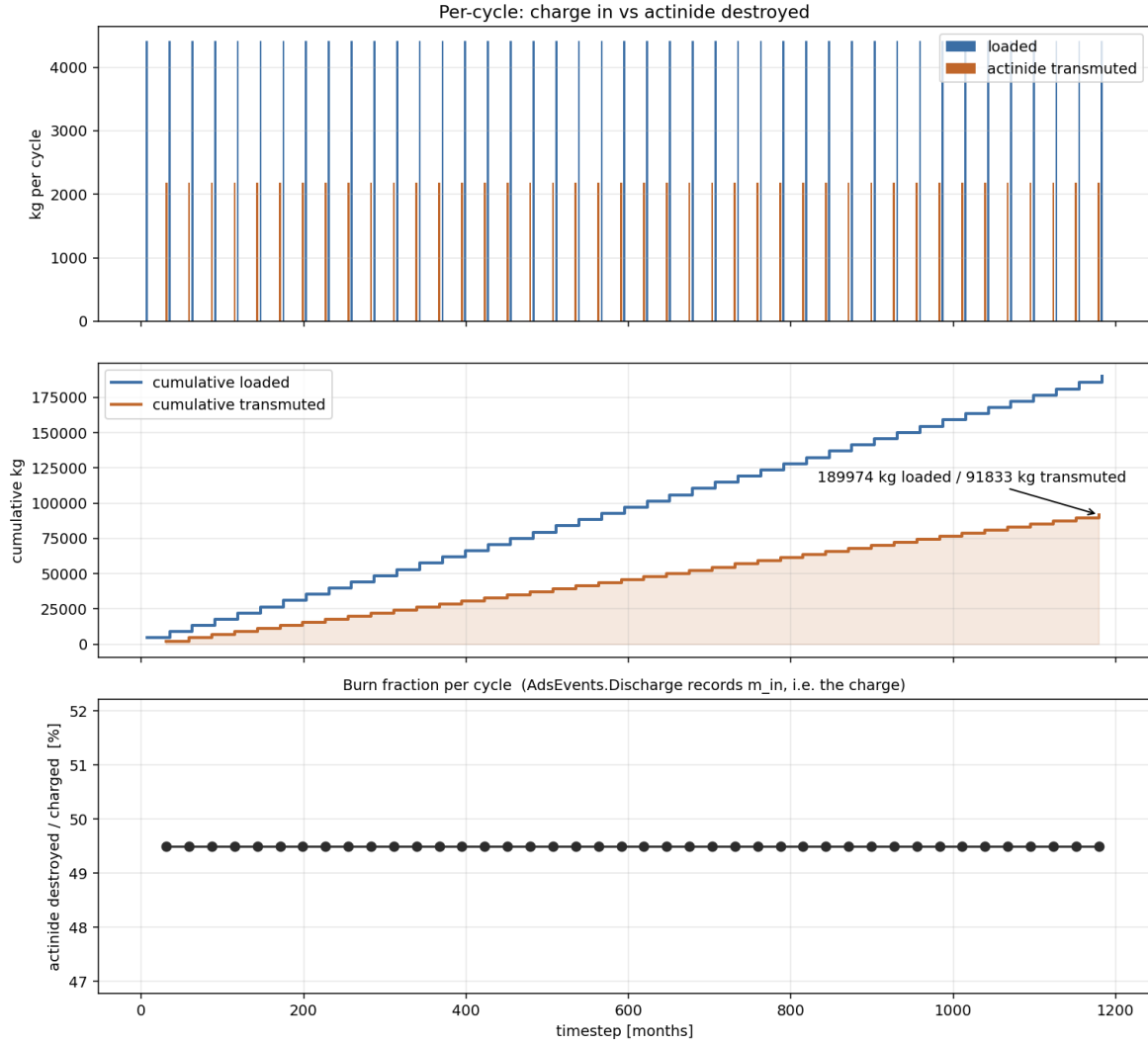


Figure 4.7: Charge and actinide destruction for each cycle of the 100-year ADS 2 recycling run. Each cycle charges 4.4 t and records 2.2 t of net actinide destruction, corresponding to a burn fraction just under 50%. The small core, which operates on a two-year cycle, completes 42 cycles over the simulation horizon.

4.3.2 Core Size and Feed Composition: the Two Burner Cases

The two burner configurations produce very different operating histories, driven entirely by core size and the resulting cycle length. Figures 4.8 and 4.9 show the core state, irradiating or idle, over the 100-year horizon for the two cases.

ADS 1 has a large core loading of 27 t and is irradiated for ten years per cycle. Over 100 years of recycling it completes only five cycles and then falls idle for the remainder of the horizon, reaching a duty factor of 50%. The core stops not because the horizon ends, but because the quantity of recoverable neptunium, americium, and curium accumulated in its feed buffer can no longer reach the required 27 t core loading. This does not mean that no actinide remains in the fuel cycle: some surviving actinide has been converted to plutonium, which is not returned to the ADS 1 feed, while additional material remains in discharged products, waste, and facility buffers. ADS 2, by contrast, has a small core loading of 4.4 t and is cycled every two years. It completes 42 cycles and runs nearly continuously across the full horizon at a duty factor of 85%. It never exhausts its feed because it consumes minor actinides slowly and supplements them with the far more abundant plutonium.

This is the central distinction between the two configurations. ADS 1 is limited by feed availability: its throughput depends on how much minor actinide the inventory can supply rather than on the physical capability of the reactor. ADS 2 is instead limited by its cycle duration. It processes material as quickly as its operating cycle allows, and the inclusion of plutonium provides enough feed to sustain that operation for a century. The tradeoff is that ADS 2 is no longer dedicated solely to minor actinide destruction. Approximately half of the material it consumes is plutonium, so it addresses the broader transuranic inventory. Over the full simulation horizon, ADS 2 destroys 91.8 t of actinides, compared with 55.8 t for ADS 1, because it burns plutonium together with the minor actinides.

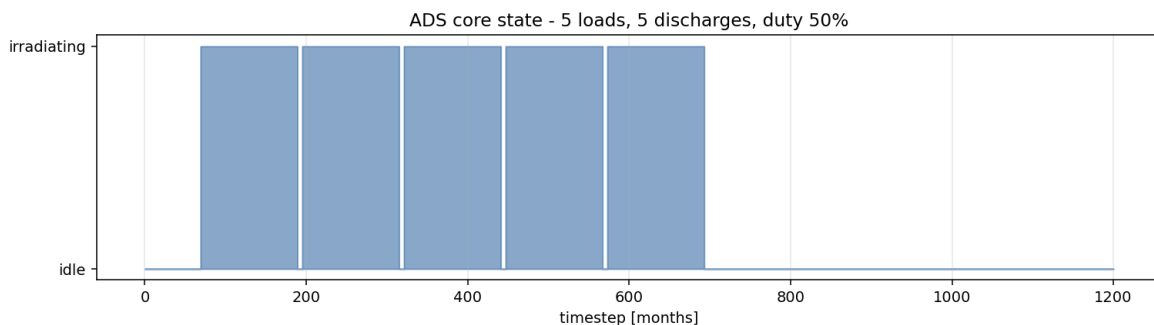


Figure 4.8: Operating state of ADS 1 over the 100-year recycling simulation, showing irradiation and idle periods. The core completes five ten-year cycles and then remains idle because the available Np, Am, and Cm feed is insufficient for another complete 27 t loading. The duty factor is 50%.

For ADS 2, the once-through and recycling runs are indistinguishable because operation is limited by the two-year cycle duration. Little surviving actinide is recovered before the next core loading is due, so recycling changes almost nothing. This result is why only one ADS 2 run is shown.

4.3.3 Recycling versus Once-Through Operation for ADS 1

ADS 1 is the only configuration in which recycling materially changes the outcome because its operation is limited by feed availability. Comparing the recycling history by cycle (Figure 4.6) with the once-through

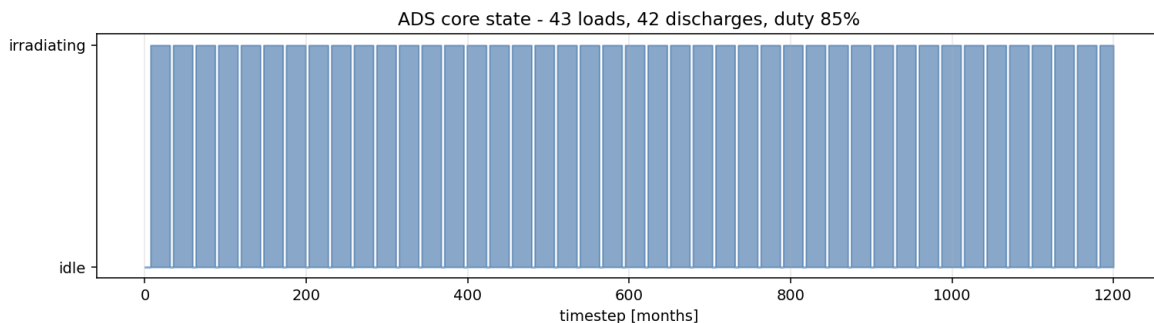


Figure 4.9: Operating state of ADS 2 over the 100-year recycling simulation, showing irradiation and idle periods. The small core, which operates on a two-year cycle, runs nearly continuously at a duty factor of 85% because plutonium provides sufficient additional feed.

history (Figure 4.10) isolates this effect. The two runs produce the same result for each completed cycle: a 27 t charge, 11.15 t of net actinide destruction, and a 41.3% burn fraction. The depletion transformation does not depend on the source of the feed. What differs is the number of cycles the burner completes before the feed is exhausted: three under once-through operation and five under recycling.

The cumulative panels make the consequence concrete. In once-through mode, the burner charges 81 t in total and records 33.5 t of net actinide destruction. With recycling, it charges 135 t and records 55.8 t, an increase of $1.67\times$ in actinide destroyed. Recycling achieves this by recovering the surviving minor actinides from each discharge and returning them to the feed, so the same atoms make more than one pass through the core and the burner sustains operation past the point where the once-through supply is spent.

The total charged mass provides the clearest evidence of reuse. The accessible minor actinide inventory is only 103.8 t (Section 4.1), yet the recycling run charges 135 t, more minor actinide material than is present in the national inventory. This is possible only because recycling returns material for additional passes; the excess of roughly 31 t represents material irradiated more than once. The once-through run, by contrast, charges 81 t, comfortably within the inventory, and simply stops when the accessible feed is gone. Even with recycling, however, the burner still stops well short of the horizon. The initial 103.8 t minor actinide inventory is only about 3.8 fresh core loadings. After each irradiation, only the surviving neptunium, americium, and curium is eligible to return to the ADS 1 feed; actinide converted to plutonium is routed outside that feed stream, and the archetype will not load a partial core. Recycling therefore extends the campaign from three cores to five, but eventually the amount of recoverable minor actinide material remains below the 27 t loading threshold.

4.3.4 Transmutation Rate and Mission Scale

Taken together, these results frame the scale of the transmutation mission. A single $3000\text{ MW}_{\text{th}}$ unit destroys actinides at up to about $1,120\text{ kg yr}^{-1}$ at full power, with the realized rate reduced in proportion to the duty factor. Over a 30-year mission, ADS 2, which runs nearly continuously, destroys on the order of 27 t of actinides per unit. ADS 1 destroys less because it stalls when its feed is exhausted. The two simulation horizons reinforce this distinction: a core limited by its cycle duration produces roughly in proportion to the mission length, whereas a core limited by feed availability plateaus after the accessible inventory is consumed. This is why extending the ADS 1 simulation from 30 to 100 years adds only two cycles.

The consequence is that sizing a fleet to destroy the minor actinide inventory is not the usual problem of

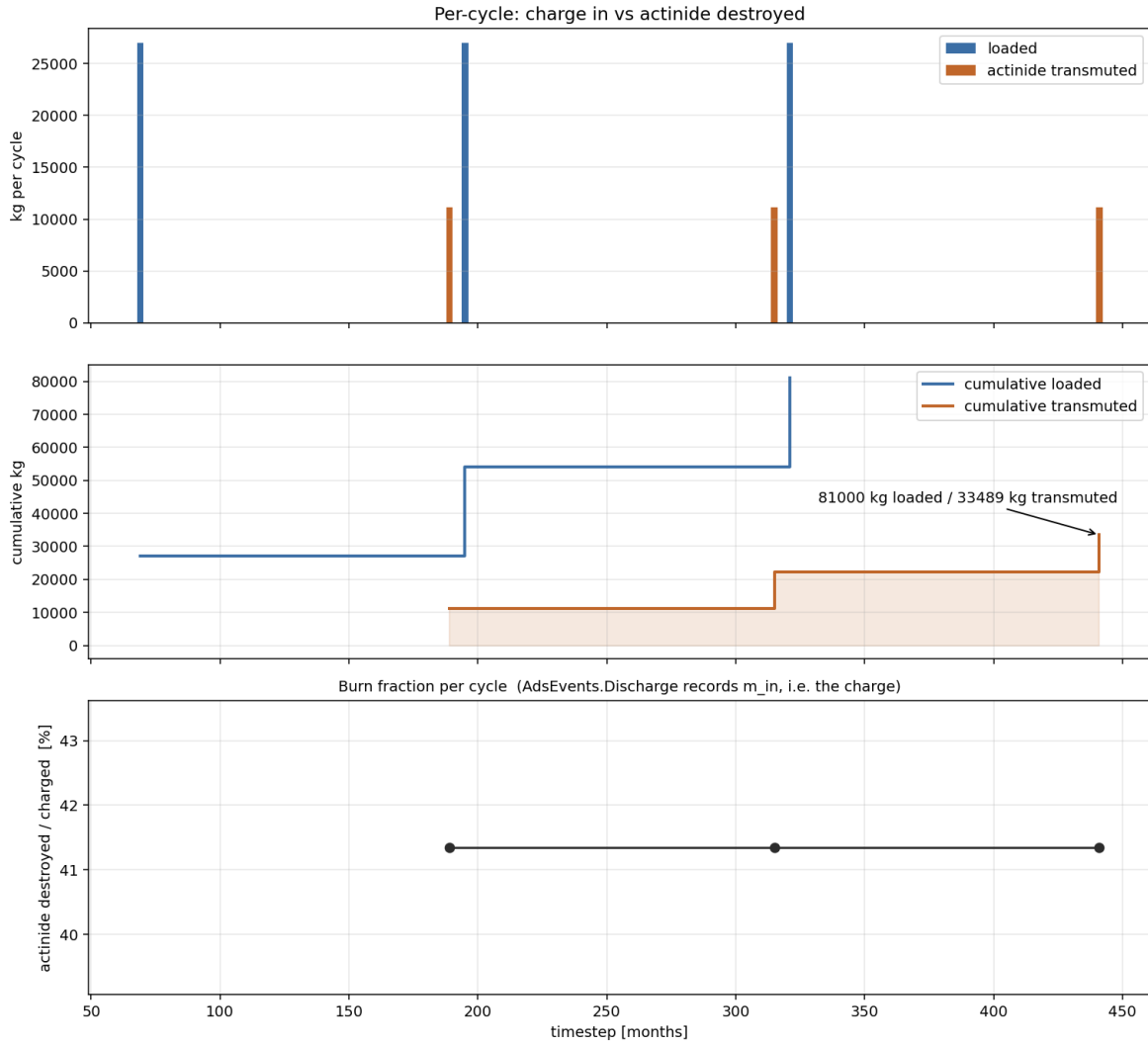


Figure 4.10: Charge and actinide destruction for each cycle of the 100-year ADS 1 once-through run. The charge, destruction, and 41.3% burn fraction for each cycle match the recycling run (Figure 4.6), but the burner completes only three cycles. It reaches 81 t charged and 33.5 t of cumulative net actinide destruction before the feed is exhausted.

determining reactor capacity. The entire national minor actinide inventory represents fewer than four core loadings. The limitation is how quickly that finite inventory can be retrieved, separated, and recycled, not how much reactor power is installed. Additional units could shorten the calendar time required to process the available feed, but they could not destroy more minor actinide material than the inventory contains. A configuration that also burns plutonium, as ADS 2 does, avoids this feed limitation but redefines the mission from minor actinide destruction to bulk transuranic destruction. This shift from a mission limited by installed power to one limited by feed availability is the central practical finding of the transmutation results and is examined further in Section 4.5.

4.4 Processed Inventory

The transmutation results of Section 4.3 describe what the burner does in each cycle; this section follows the material through the whole fuel cycle and asks what becomes of the national inventory once it is processed. Every kilogram drawn from `us_inventory` meets one of three fates: it is destroyed by fission in the ADS, it is recovered and returned to the feed under recycling, or it is routed to disposal. Tracking these streams shows both what the mission accomplishes and what it leaves behind, and connects the results back to the waste management motivation of Chapter 2.

4.4.1 Disposition of the Minor Actinide Inventory

Table 4.1 closes the actinide mass balance against the 103.8 t of minor actinides initially present in the accessible inventory (Section 4.1). The quantity labeled “undestroyed descendant actinide” is the initial actinide mass minus the net actinide destruction recorded by the ADS. It is therefore not equivalent to feed of minor actinides still available to the burner: it includes any surviving neptunium, americium, and curium, actinides produced by transmutation such as plutonium, material routed to waste, and material held in facility buffers.

In once-through operation the burner records 33.5 t of net actinide destruction and leaves 70.3 t of descendant actinide undestroyed. Of that remainder, roughly 47.5 t is surviving actinide discharged to waste, while about 22.8 t remains in unprocessed inventory because it is insufficient to form another 27 t core. With recycling, net actinide destruction increases to 55.8 t, leaving 48.0 t of descendant actinide somewhere in the modeled fuel cycle. The 48.0 t value does not imply that another 27 t ADS 1 core could be loaded. In the ADS 1 flowsheet, only Np, Am, and Cm are recovered as burner feed; plutonium produced during irradiation is routed outside the minor actinide feed stream, and the remaining recoverable minor actinide inventory is distributed among products and buffers. The ADS therefore stalls when its feed buffer cannot again accumulate a full 27 t loading.

Table 4.1: Actinide mass balance relative to the 103.8 t of minor actinide initially present in the accessible inventory at the end of the 100-year ADS 1 runs. “Undestroyed descendant actinide” includes surviving minor actinides, actinides produced by transmutation, material routed to waste, and material held in facility buffers; it is not the mass of usable ADS 1 feed remaining.

Disposition	Once-through	Recycle
Net actinide destruction	33.5 t	55.8 t
Undestroyed descendant actinide	70.3 t	48.0 t
Initial accessible MA inventory	103.8 t	103.8 t

4.4.2 Representative ADS Discharge Compositions

Figure 4.11 shows the isotopic composition of one representative 27 t discharge from ADS 1. The composition is extracted from the ADS output received by the downstream separations facility before that material is partitioned or combined with another stream. It therefore excludes the uranium, plutonium, and fission products removed from the original spent fuel during front-end separation. This distinction is important: the previous repository inventory analysis combined the ADS discharge with the much larger front-end separations reject stream and did not represent the material leaving the ADS alone.

The discharge contains surviving neptunium, americium, and curium, together with fission products and actinides generated during irradiation. Although the initial ADS 1 loading contains only neptunium, americium, and curium, plutonium isotopes, particularly ^{238}Pu , appear prominently in the discharge. This plutonium is not material carried into the core from the front-end reject stream. The discharged composition is obtained from the full SERPENT depletion inventory, in which neutron-induced transmutation and radioactive decay chains convert part of the initial minor actinide inventory into descendant plutonium isotopes and small quantities of uranium. These products remain part of the discharged actinide inventory and are therefore not counted as destroyed actinide.

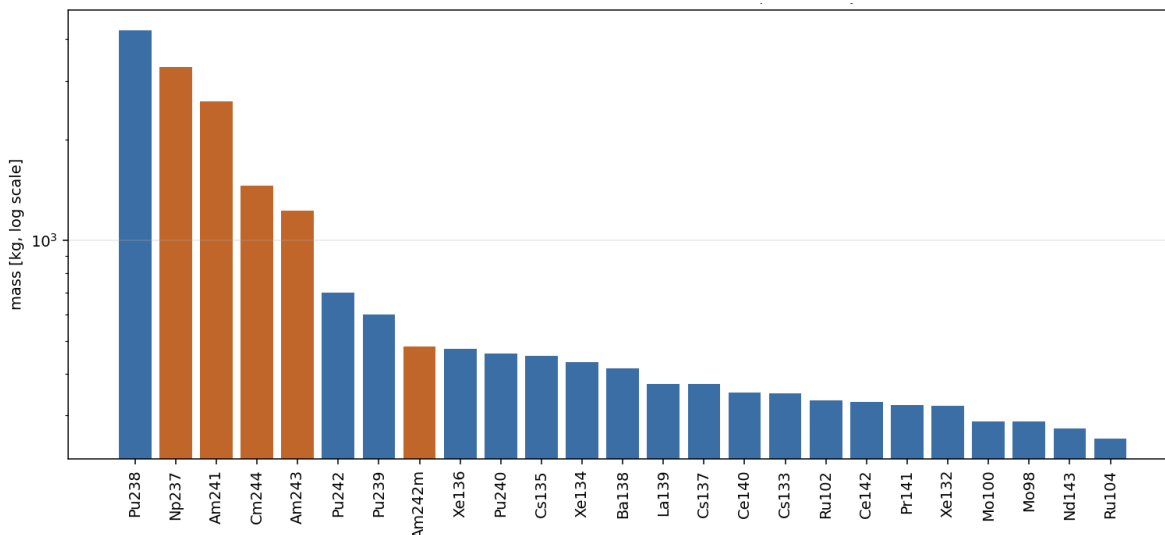


Figure 4.11: Isotopic composition of one representative 27 t discharge from ADS 1, showing the 25 most abundant nuclides on a logarithmic mass scale; minor actinides are highlighted. The composition is extracted from the ADS output before downstream separation and excludes the front-end separations reject stream. Although the initial core contains only neptunium, americium, and curium, plutonium isotopes and small quantities of uranium appear as descendant actinides produced during irradiation in the SERPENT depletion calculation.

Figure 4.12 shows one representative discharge from ADS 2, extracted using the same ADS-only method. Unlike ADS 1, the ADS 2 core is initially loaded with approximately equal masses of plutonium and minor actinides. Plutonium isotopes therefore remain the largest actinide components of the discharge, followed by surviving and produced neptunium, americium, and curium. The stream also contains the fission products generated during the irradiation cycle.

Under recycling, the downstream separations facility recovers plutonium, neptunium, americium, and curium from this discharge and returns those actinides to the burner feed. Fission products and other

non-actinides are instead removed from the recycle stream and routed to waste. ADS 2 therefore retains plutonium within the burner cycle rather than routing it directly outside the burner feed, which allows it to address the broader transuranic inventory rather than minor actinides alone.

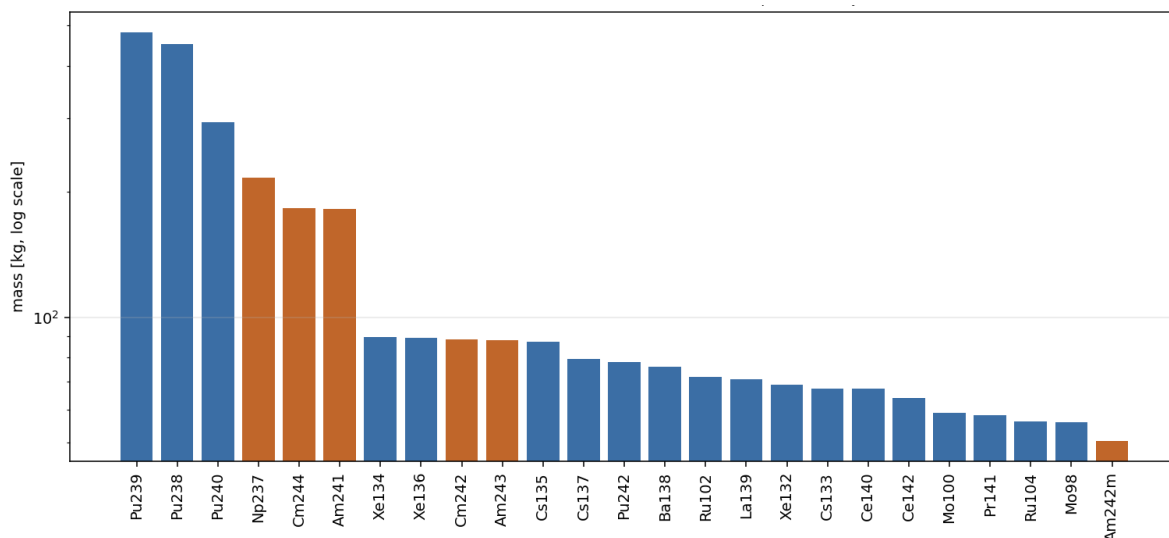


Figure 4.12: Isotopic composition of one representative ADS 2 discharge, showing the 25 most abundant nuclides on a logarithmic mass scale; minor actinides are highlighted. The stream is extracted directly from the ADS output before downstream separation. Plutonium is the largest actinide component, followed by minor actinides, while the discharge also contains fission products generated during irradiation. Under recycling, plutonium and the minor actinides are recovered for reuse, while fission products and other non-actinides are routed to waste.

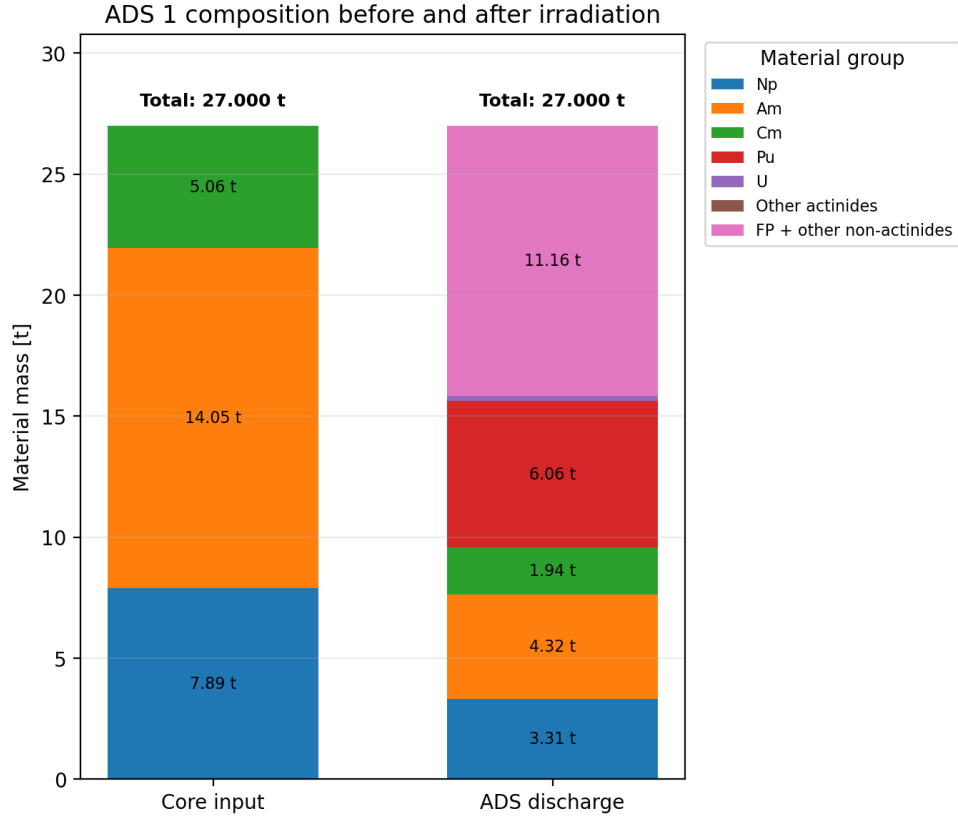
4.4.3 Representative Core Input–Output Comparison

The isotope-specific discharge plots in

Figures 4.11 and 4.12 identify the principal nuclides leaving each burner. To assess the effectiveness of the transmutation step more directly, Figures 4.13 and 4.14 compare the material entering and leaving one matched irradiation cycle for each configuration. Each comparison represents one core loading and its corresponding discharge, rather than the cumulative throughput of the 100-year simulation. This per-cycle basis avoids counting recycled material more than once and isolates the composition change produced by the ADS itself.

For ADS 1, the total actinide inventory decreases from 27.00 t in the initial core to approximately 15.84 t in the discharge. The remaining mass appears primarily as fission products and other non-actinides. The redistribution among the actinide groups is also significant: neptunium, americium, and curium decrease, while plutonium is produced. The decrease of an individual minor actinide group is therefore larger than the net actinide destruction because part of that material is converted into another actinide rather than fissioned.

ADS 2 reduces the actinide mass from 4.418 t to approximately 2.231 t per cycle, corresponding to a 49.5% net reduction. Its larger fractional reduction reflects the plutonium-bearing feed, but the discharge still contains substantial plutonium and minor actinides that require recovery and additional irradiation for further destruction. These comparisons show that neither burner eliminates its actinide loading in



Matched cycle: load at timestep 573; discharge at timestep 693

Figure 4.13: Stacked comparison of the material entering and leaving one representative ADS 1 irradiation cycle. The 27 t input contains neptunium, americium, and curium. The discharge contains approximately 15.84 t of remaining actinides and 11.16 t of fission products and other non-actinides. The net actinide reduction is approximately 11.16 t, or 41.3% of the input actinide mass. Plutonium and the small uranium component in the discharge are descendant actinides produced during irradiation.

one pass. Instead, each converts approximately 41–50% of the loaded actinide mass per cycle, making the availability and performance of the downstream separations and recycle system central to the overall mission.

Taken together, these results qualify what transmutation achieves for the national inventory. Over a century, even the more favorable recycling configuration destroys only about half of the accessible minor actinides. Achieving that result requires processing essentially the entire spent fuel inventory. Most of the processed material, including uranium, fission products, and, for ADS 1, plutonium, still requires disposal. The mission changes the isotopic character of the waste by removing minor actinides, which contribute strongly to its long-term radiotoxicity and heat load, but it does not substantially reduce the bulk mass requiring management and disposal. Quantifying the resulting reduction in radiotoxicity or decay heat would require coupling these mass histories to a decay and dose calculation. That analysis is beyond the scope of this work but is a natural extension discussed in Chapter 5. The fuel cycle results instead establish the upstream reality against which any such benefit must be weighed: minor actinide transmutation is constrained not by reactor performance but by the scarcity of its feed and the scale of the inventory that must be handled to recover it.

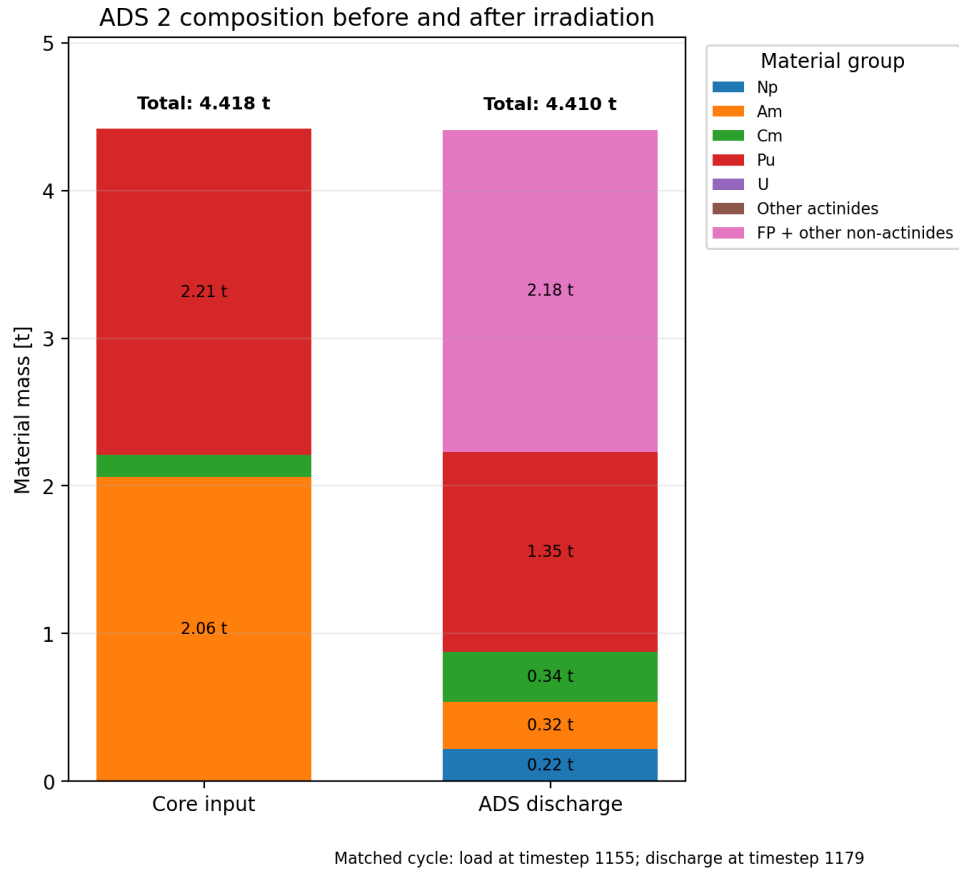


Figure 4.14: Stacked comparison of the material entering and leaving one representative ADS 2 irradiation cycle. The 4.418 t input contains approximately equal masses of plutonium and minor actinides. The discharge contains approximately 2.23 t of remaining actinides and 2.18 t of fission products and other non-actinides. The net actinide reduction is approximately 2.19 t, or 49.5% of the input actinide mass. The small difference between the total input and output masses is the mass defect recorded by the reduced-order depletion model.

4.5 Discussion

The results of this chapter can be read as a single argument about scale. The `us_inventory` and `ads` archetypes were built to connect a detailed, assembly-resolved picture of the U.S. spent fuel inventory to a reduced-order model of an accelerator-driven burner, and running them together turns qualitative expectations about minor actinide transmutation into quantified constraints. Two findings dominate, and both point to the same conclusion: the scale and dilution of the national inventory, not the capability of the burner, set the limits of what transmutation can achieve. The first is that the mission is limited by the availability of its minor actinide feed rather than by installed reactor capacity. The second is that, because that feed is such a small fraction of the spent fuel, the mission changes the composition of the waste far more than its quantity. In each case the binding factor lies in the fuel cycle required to retrieve, separate, and handle a large, dilute inventory to reach a scarce target material, not in the accelerator-driven system that consumes it.

4.5.1 The Mission is Feed-Bound

The central result is that a minor actinide transmutation mission is limited by the availability of its feed rather than by installed reactor capacity. The accessible minor actinide inventory is 103.8 t, fewer than four loadings of a single 27 t core. The binding question is therefore not how much power can be deployed, but how quickly this finite, dilute inventory can be retrieved, separated, and recycled. ADS 1 demonstrates this limitation: it stalls after three cores under once-through operation or five with recycling (Section 4.3.3). Increasing reactor power cannot change that outcome because sufficient feed for additional cores does not exist. This inverts the usual framing of reactor deployment, in which capacity is the constraint and fuel is assumed abundant.

A simple rate calculation frames the fleet question that follows from this. A 3000 MW_{th} unit destroys actinide at up to about 1,120 kg yr⁻¹ at full power (Section 4.3.4), so clearing the 103.8 t inventory represents on the order of 90 unit-years of operation at full power. Distributed over a 30-year mission, this is roughly three units' worth of destruction capacity. A single unit would require most of a century, and ADS 1 could not operate at full duty for that long because of limited feed availability. This is not a precise fleet sizing estimate because it omits reprocessing losses, retrieval and transportation logistics, and the reduction in duty factor imposed by feed availability. It nevertheless establishes the order of magnitude: the transmutation of the national minor actinide inventory would require a handful of high-power units operating over several decades, and its pace would be set by fuel handling rather than by the number of units constructed.

ADS 2 avoids the feed limitation, but only by changing the mission. Plutonium is approximately ten times more abundant in the inventory than minor actinides, and burning it together with minor actinides provides enough feed for the small core to operate for the full century at an 85% duty factor. ADS 2 destroys 91.8 t of actinides, compared with 55.8 t for ADS 1 (Section 4.3.2). However, roughly half of the ADS 2 feed is plutonium, so it functions as a transuranic burner rather than a dedicated minor actinide burner. The choice between the configurations is therefore a choice of mission: destroy scarce minor actinides with high radiotoxicity at a rate bounded by their availability, or destroy the broader transuranic inventory at a higher rate by accepting plutonium as feed.

4.5.2 Transmutation Changes Waste Composition

The second finding concerns what the mission does to the waste. The representative input–output comparisons in Figures 4.13 and 4.14 show that neither burner eliminates its actinide loading in one irradiation pass. ADS 1 reduces the loaded actinide mass by 41.3% per cycle, while ADS 2 reduces it by 49.5%. The remaining actinides require further separation and recycle if additional destruction is desired.

These per-cycle reductions describe the performance of the burner itself, but they do not translate into a comparable reduction in the total mass requiring disposal. Minor actinides constitute only 0.12% of the spent fuel, so recovering the approximately 100 t accessible inventory requires separations to process on the order of the entire drawn spent fuel inventory. Uranium and fission products removed during front-end separation therefore dominate the disposal stream. In the ADS 1 fuel cycle, plutonium produced from the minor actinide feed is also routed outside the ADS 1 recycle stream. The principal modeled effect of transmutation is consequently a change in isotopic composition through the destruction and redistribution of actinides, rather than a substantial reduction in the bulk mass routed to waste (Section 4.4).

Repository volume, requirements for waste packages, and effects on the disposal footprint were not evaluated in this work. The modeled transmutation process removes minor actinides, which contribute strongly to the long-term radiotoxicity and decay heat of spent fuel. This is a meaningful outcome for repository performance, but it changes the composition of the waste rather than substantially reducing its mass, and it requires handling the full inventory.

This reframes the value proposition of transmutation in a way the fuel cycle model is well suited to expose. The benefit is not a reduction in waste mass, but the removal of long-lived constituents. The cost is a fuel cycle that must retrieve, transport, and reprocess essentially the entire national inventory to recover a trace component. Whether that trade is worthwhile is a question about repository requirements and reprocessing capacity, not about the burner itself.

4.5.3 Limitations

Several limitations constrain the interpretation of these results, and identifying them also marks the most useful directions for further work.

The ADS model is reduced-order. The burner does not solve neutron transport during the simulation; it reads precomputed SERPENT depletion cases, selects the nearest case based on power and input composition, and interpolates along the selected case’s irradiation time axis (Section 3.4). This approach is appropriate for a fuel cycle study, where the quantity of interest is the mass transmuted per core rather than the detailed physics within the core. The destruction rate for each cycle was shown to agree with an independent fission energy balance to within a few percent (Section 4.3.1). The model nevertheless inherits the assumptions of the underlying cases, including a single homogeneous fuel region, a fixed geometry, and a library spanning only the two representative feeds used here. It cannot capture behavior outside the tabulated cases. A core whose composition drifts far from any library case is matched to the nearest one rather than modeled directly. The constant burn fraction from cycle to cycle results from irradiating every core to the same point on the same trajectory, not a physical invariant.

The inventory selection results characterize the feed directly from the static inventory rather than from the coupled simulation (Section 4.2). This treatment is exact because the selection is a deterministic ordering of a fixed pool. However, the policy comparison isolates the supply side and does not reflect any feedback between selection and burner operation. All production scenarios use the single `highest_minor_actinide`

policy, so the effect of policy on outcomes for the full fuel cycle, rather than only on the feed, is left for future work.

A more fundamental system level limitation is that the enabling separations infrastructure is assumed rather than evaluated. Both burner configurations require the national spent fuel inventory to be reprocessed before the ADS can receive its feed, and ADS 2 additionally requires plutonium and minor actinides to be recovered as separate streams and blended in a prescribed ratio. The technical readiness, recovery losses, product quality, safeguards requirements, cost, and policy acceptability of such an infrastructure may be at least as consequential to deployment as the performance of the burner itself. The present simulations therefore should not be interpreted as demonstrating the feasibility of the complete transmutation system; they evaluate its material flows after assuming that the necessary separations capability exists.

The study also does not include a reprocessing only reference case. It therefore cannot isolate how much of a potential repository benefit would arise from partitioning uranium, plutonium, minor actinides, and fission products into separate streams, and how much additional benefit would result specifically from irradiation in the ADS. That distinction would require applying the same decay heat, radiotoxicity, and disposal analysis to an unprocessed case, a reprocessing only case, and the reprocessing plus transmutation cases.

Finally, the study stops at the mass balance. It quantifies what is destroyed and what remains but does not evaluate the radiological or economic consequences that determine whether the mission is worthwhile. Radioactive decay is modeled within the simulation, so material compositions are aged as they move through the fuel cycle. However, no downstream radiotoxicity or decay heat calculation is applied to the final waste compositions, so the reduction in long-term radiological hazard that motivates minor actinide removal is not quantified here. Likewise, no cost, facility, or transportation models accompany the fuel cycle, so the substantial reprocessing and handling burden implied by the results is described but not priced. The model also does not represent the retrieval and transportation logistics for which the provenance results (Section 4.2.2) provide useful inputs.

These omissions identify important directions for future work. The assembly-resolved provenance records, mass-conserving discharge streams, and operating histories developed here provide the inputs required for subsequent radiotoxicity, economic, transportation, and siting analyses. These extensions are discussed further in Chapter 5.

Chapter 5

Conclusions and Future Work

This thesis set out to connect a detailed picture of the U.S. used fuel inventory to a model of an accelerator driven burner, so that the transmutation of minor actinides could be evaluated as a national fuel cycle rather than as an isolated reactor calculation. Two cyclis archetypes were developed to that end. `us_inventory` represents the inventory at assembly resolution, supplying material by a choice of selection policies while preserving each assembly’s composition and provenance; `ads` models an accelerator driven minor actinide burner from precomputed Serpent depletion cases, transforming a loaded core into its irradiated composition and closing the actinide mass balance each cycle. Together they form a complete transmutation fuel cycle when combined with standard separations, mixing, and disposal facilities.

Exercising this fuel cycle across eight scenarios produced two principal findings. First, minor actinide transmutation is limited by the availability of its feed rather than by ADS capacity. The accessible minor actinide inventory is approximately 103.8 t, fewer than four fresh loadings of the 27 t ADS 1 core. ADS 1 therefore stalls after three cycles under once through operation and five cycles with recycling. ADS 2 avoids this limitation only by coburning plutonium, thereby changing the mission from dedicated minor actinide destruction to broader transuranic destruction.

Second, the direct input–output comparisons show that neither ADS configuration eliminates its loaded actinides in one irradiation pass. ADS 1 reduces the actinide mass from 27.00 t to approximately 15.84 t, corresponding to 11.16 t, or 41.3%, of net actinide destruction per cycle. ADS 2 reduces the actinide mass from 4.418 t to approximately 2.231 t, corresponding to 2.187 t, or 49.5%, of net destruction per cycle.

The remaining actinides must be recovered and irradiated again if greater destruction is required. At the full fuel cycle scale, the 100 year ADS 1 recycling scenario destroys 55.8 t of the initial 103.8 t accessible minor actinide inventory, while the once through case destroys 33.5 t.

These results show that accelerator driven transmutation is not a stand alone solution to the spent fuel problem. It is a downstream component of a much larger reprocessing and recycle system. Before the burner can provide any benefit, essentially the entire selected spent fuel inventory must be retrieved and processed to recover the comparatively small plutonium and minor actinide fractions. The simulations assume that these materials can be recovered into separate streams at high efficiency; they do not demonstrate the technical, economic, safeguards, licensing, or policy feasibility of that infrastructure. The difficulty of establishing and operating the required reprocessing system may therefore be at least as important as the difficulty of designing the ADS itself.

Partitioning also changes the composition and management of the waste streams before any ADS irradiation.

tion occurs. Separating uranium, fission products, plutonium, and minor actinides may itself alter decay heat, radiotoxicity, waste form requirements, and repository performance, depending on the disposition of each stream. Because this thesis does not include a reprocessing only reference scenario or a downstream radiotoxicity calculation, it cannot determine how much of the potential long term waste benefit would result from partitioning alone and how much would result from the additional actinide destruction in the ADS. The contribution quantified here is therefore the incremental net destruction performed by the burner after separation: 41.3% per pass for ADS 1 and 49.5% for ADS 2.

Several extensions follow directly from this work. The most immediate is to couple the mass-conserving waste histories to a decay and dose calculation so that the reductions in long term radiotoxicity and decay heat, which motivate minor actinide removal, can be quantified rather than inferred. That analysis should include a reprocessing only reference scenario in which uranium, plutonium, minor actinides, and fission products are partitioned but not irradiated in an ADS. Comparing that case with both the unprocessed inventory and the ADS scenarios would distinguish the repository benefit of separation itself from the incremental benefit of transmutation.

A second extension is to use the assembly-level provenance recorded by `us_inventory`, which associates each supplied assembly with its source facility and coordinates, as the input to a transportation and siting analysis of where retrieval and processing would occur. A third is to develop an economic model of the reprocessing and handling burden implied by the results, which this study describes but does not price. Finally, the reduced order `ads` model could be broadened with a larger library of depletion cases, and the effects of inventory selection policies could be examined within the fully coupled simulation rather than only against the static inventory. In each case, the archetypes and results developed here provide a foundation for these subsequent analyses.

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